Natural boundaries of the full bulk susceptibility for the fixed-ratio anisotropic square-lattice Ising model

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Abstract

For each fixed positive coupling ratio, we determine the natural boundaries of the high-temperature bulk susceptibility and the low-temperature connected pure-phase susceptibility. The continuation domains in the complex temperature variable are a bounded lens and a simply connected epigraph, respectively. At resonant edges, a nonzero leading derivative survives the differentiated particle tail. At negative high-temperature edges the odd momentum constraint excludes endpoint resonance. There a symmetric pole normal form and a dimension-uniform beta-period estimate identify a single dominating particle index at selected Taylor orders. The isotropic case follows by matching the full-lattice coefficients and prefactors to the normalized isotropic theorems.

Introduction

The form-factor expansion expresses the magnetic susceptibility of the square-lattice Ising model as an infinite sum of multiple integrals. The singularities of the individual integrals and their possible cancellation in the complete sum are central to its complex-temperature analyticity. Orrick, Nickel, Guttmann and Perk [3, Section 3.2] studied the order dependence of the isotropic singularity amplitudes. Boukraa, Hassani and Maillard [4, Section 6.2] described the anisotropic cosine-grid singularity equations in two complex couplings and their accumulation. Fixing the physical coupling ratio gives a one-variable problem in which the physical components and their boundary singularities can be determined explicitly.

The published spin form factors of Bugrij and Lisovyy [1] and Iorgov and Lisovyy [2] identify the coefficients of the two bulk susceptibilities in Theorem 1.1. At resonant edges the proof uses the first singular derivative and an absolute tail estimate. At negative high-temperature edges it uses the growth of boundary Taylor coefficients. The equal-coupling case uses the isotropic natural-boundary theorems of Lin [7, 8] via the coefficient identities in Section 6.

AI Declaration

ChatGPT was used extensively in the mathematical research, proof development, and preparation of this manuscript.

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1 Fixed-ratio slice and physical susceptibility

Fix $J_x > 0$ and $\lambda = J_y/J_x \in (0, \infty)$. Throughout,

$$z = 2\beta J_x, \qquad S = \sinh z, \quad T = \sinh(\lambda z), \quad A = \cosh z \cosh(\lambda z), \quad p = ST. \tag{1.1}$$

The coupling ratio λ is held fixed in the complex continuation, during which T/S generally varies. There is a unique $z_c > 0$ with $\sinh z_c \sinh(\lambda z_c) = 1$. Define strictly increasing functions on $[0, \infty)$ by

$$P_\lambda(x) = \cosh x \tanh(\lambda x), \qquad Q_\lambda(x) = \cosh(\lambda x) \tanh x. \tag{1.2}$$

Their values at z_c are both one. Set

$$Y_\lambda = \frac{\pi}{2 \max(1, \lambda)},$$

$$H_\lambda(y) = \min\{P_\lambda^{-1}(\cos y), Q_\lambda^{-1}(\cos(\lambda y))\}, \qquad |y| < Y_\lambda, \tag{1.3}$$

$$L_\lambda(y) = \max\{P_\lambda^{-1}(|\cos y|), Q_\lambda^{-1}(|\cos(\lambda y)|)\}, \qquad y \in \mathbb{R}. \tag{1.4}$$

The domains of interest are specified independently of susceptibility:

$$\Omega_H = \{x + \mathrm{i}y : |y| < Y_\lambda,\ |x| < H_\lambda(y)\}, \qquad \Omega_L = \{x + \mathrm{i}y : x > L_\lambda(y)\}. \tag{1.5}$$

The value of H_λ at each endpoint is its continuous limit zero.

Theorem 1.1 (Full bulk susceptibility at every positive ratio)**.** *Let $0 < \lambda < \infty$. The high-temperature physical bulk susceptibility germ extends holomorphically to Ω_H, and $\partial\Omega_H$ is its natural boundary. The low-temperature physical pure-phase connected bulk susceptibility germ extends holomorphically to Ω_L, and the entire graph $\partial\Omega_L = \{L_\lambda(y) + \mathrm{i}y : y \in \mathbb{R}\}$ is its natural boundary. Here natural boundary means that no boundary point has a neighborhood carrying a holomorphic function agreeing with the indicated continuation on its domain side. Both statements concern the entire lattice and all particle indices. They apply separately to $\mathcal{X} = \beta^{-1}\chi$ and to χ with the field convention below.*

For $\lambda \neq 1$, Sections 2–5 give the local boundary estimates with constants depending on the fixed ratio and selected boundary point. These constants can degenerate as $\lambda \to 1$. At $\lambda = 1$ the proof uses the normalized isotropic theorems [7, 8]. Section 6 gives the full-site coefficient identities and the covering relations for this application.

1.1 Field normalization and state

Let the magnetic field have energy units, so the finite-volume Hamiltonian contains $-h \sum_v \sigma_v$, with $\sigma_v \in \{-1, 1\}$ and magnetic moment set to one. Direct differentiation of the finite partition function gives

$$\left.\frac{\partial \langle \sigma_0 \rangle}{\partial h}\right|_{h=0} = \beta \sum_v \big(\langle \sigma_0 \sigma_v \rangle - \langle \sigma_0 \rangle \langle \sigma_v \rangle\big).$$

Thus the normalized response is the covariance sum $\mathcal{X}$, and $\chi = \beta\mathcal{X}$. A dimensionless field $\eta = \beta h$ instead has response $\mathcal{X}$. A specified nonunit magnetic moment multiplies both by its square.

On $0 < z < z_c$ the Gibbs state is unique and its magnetization is zero:

$$\mathcal{X}_H(z) = \sum_{(m,l) \in \mathbb{Z}^2} \langle \sigma_{00} \sigma_{ml} \rangle.$$

On $z > z_c$ take the thermodynamic plus state, or equivalently the minus state for this connected observable:

$$\mathcal{X}_L(z) = \sum_{(m,l)\in\mathbb{Z}^2} (\langle\sigma_{00}\sigma_{ml}\rangle_+ - M^2), \qquad M^2 = (1-p^{-2})^{1/4}.$$

The fourth root is positive on the displayed real interval. For the energy-field response in this phase, the derivative is the zero-field pure-phase linear response, equivalently the limit from $h > 0$ in the plus phase (or from $h < 0$ in the minus phase). The covariance formula is obtained with that phase fixed before taking the zero-field limit. A symmetric mixture has the same two-spin expectation but zero one-spin mean; using its zero mean as the subtraction would leave the nondecaying term M^2 and would define a different, divergent sum. This distinction is retained through the thermodynamic form-factor limit below.

Once the absolute summability proved below is available, two inequalities identify the covariance sum with the thermodynamic uniform-field derivative. Finite-volume concavity gives the upper bound; a field supported on a finite set gives the lower bound. The GHS inequality for arbitrary nonnegative couplings and site fields [6, Theorem 1.5] says that truncated two-spin correlations decrease as any nonnegative site field increases. Pinning the spins removed from a larger plus box by fields tending to $+\infty$ consequently shows that the plus-box covariances increase with the box. Their sums converge monotonically to $\mathcal{X}$. If $m_V(h) = \langle\sigma_0\rangle_{V,+,h}$, GHS also makes m_V concave for $h \geq 0$. Thus

$$m_V(h) - m_V(0) \leq h m_V'(0) = \beta h \sum_{v\in V} \mathrm{Cov}_{V,+,0}(\sigma_0, \sigma_v) \leq \beta h \mathcal{X}.$$

Passing to the plus-state limit gives the upper right-derivative bound $m'(0+) \leq \beta\mathcal{X}$. Conversely, turn on h only in a finite set E in the infinite plus state. This is a bounded local change of measure, so its derivative at zero is exactly $\beta\sum_{v\in E}\mathrm{Cov}_{+,0}(\sigma_0,\sigma_v)$. Ferromagnetic field monotonicity makes the uniform-field magnetization at least this local-field magnetization, with equality at $h = 0$. Letting E increase to the lattice gives the reverse derivative bound. Hence $m'(0+) = \beta\mathcal{X}$. In HT, spin-flip symmetry gives the same derivative from the other side; in LT the stated one-sided pure-phase convention applies. The two bounds determine the derivative after the thermodynamic limit; absolute summability controls the passage from finite covariance sums to the whole-lattice sum.

1.2 Real-temperature spin form factors

Away from criticality on either positive real interval set

$$B = T/S > 0, \qquad C = A/S, \qquad \cosh\gamma(q) = W(q) = C - B\cos q, \qquad \gamma(q) > 0.$$

Let $f_N(m,l)$ be

$$\frac{B^{N^2/2}}{N!}\int_{\mathbb{T}^N}\prod_{i=1}^N \frac{e^{-|m|\gamma(q_i)+\mathrm{i}lq_i}}{\sinh\gamma(q_i)}\prod_{i<j}\left[\frac{\sin((q_i-q_j)/2)}{\sinh((\gamma(q_i)+\gamma(q_j))/2)}\right]^2\prod_{i=1}^N\frac{\mathrm{d}q_i}{2\pi}. \tag{1.6}$$

The infinite-plane correlations are

$$\langle\sigma_{00}\sigma_{ml}\rangle = \xi_H \sum_{N\geq 1,\ N \text{ odd}} f_N(m,l), \qquad \xi_H = \frac{(1-p^2)^{1/4}}{\sqrt{p}}, \tag{1.7}$$

$$\langle\sigma_{00}\sigma_{ml}\rangle_+ = \xi_L\left(1 + \sum_{N\geq 2,\ N \text{ even}} f_N(m,l)\right), \qquad \xi_L = (1-p^{-2})^{1/4} = M^2. \tag{1.8}$$

All powers are fixed by their positive real values. These are the anisotropic vacuum-to-state spin form factors of [1, 2], in the bulk normalization of [3], with the factor $B^{N^2/2}$ in the Nth term.

For the plane limit use the finite-cylinder formulas at arbitrary separation in [1, (35)–(39)]. The vacuum-to-state matrix elements are given in [2, Theorem 6.1] in the LT phase and in its Remark 6.2, equations (6.8)–(6.10), with the HT vacuum change. On a fixed real noncritical compact set, the one-body dispersion is analytic in a common angular strip and has a positive real gap. The finite-circumference correction functions η, Λ^{-1} and $\log \xi_T$ tend uniformly to zero: their sum-minus-integral expressions are controlled by the exponentially convergent trapezoidal rule for periodic analytic functions. They are uniformly bounded before the limit.

The Schur contraction gives a summable majorant for the particle sum in the cylinder limit. For real roots $0 < \rho(q) = e^{-\gamma(q)} \le r < 1$, the Schur factor $|(\rho_i - \rho_j)/(1 - \rho_i\rho_j)|$ is at most a constant strictly below one. The matched-Pfaffian argument proved in Section 3 applies to the discrete angular grids as an algebraic identity, with their normalized counting measures. Its single-body finite-size corrections cost only C^N. Consequently the finite-circumference coefficients, at each fixed separation, have a common bound $\exp(-cN^2 + O(N\log(N+1)))$, including their factorial normalization. This is summable independently of circumference, so the particle sum passes through the cylinder limit; taking each fixed-N Riemann-sum limit gives (1.6).

Every local cylinder limit is a translation-invariant, spin-flip invariant Gibbs state for these finite-range positive couplings. Apply [5, Theorem 1] with the vertex-transitive subgroup consisting of translations: unequal horizontal and vertical couplings satisfy all four stated hypotheses. Such a state is the equal mixture of the plus and minus states. Its even spin correlations are those of the plus state. This identifies the two-point function in (1.8); the connected pure-phase observable is obtained by removing the zero-particle term ξ_L.

1.3 Symmetric double contours and every lattice site

The symmetric contour represents the two lattice directions by independent variables and turns the lattice sum into a rational multiplier. For separate contour variables x, y put

$$D(x,y;z) = A - \frac{S}{2}(x + x^{-1}) - \frac{T}{2}(y + y^{-1}), \qquad \mathscr{A}(v,w) = \frac{v-w}{1-vw}. \tag{1.9}$$

With $X = \prod_i x_i$ and $Y = \prod_i y_i$, the same correlation coefficient is

$$\frac{p^{N/2}}{N!} \oint \cdots \oint X^{|m|-1} Y^{|l|-1} \prod_{i<j} \mathscr{A}(x_i,x_j)\mathscr{A}(y_i,y_j) \prod_i \frac{\mathrm{d}x_i\, \mathrm{d}y_i}{(2\pi\mathrm{i})^2 D(x_i,y_i;z)}. \tag{1.10}$$

The circles are positively oriented and sufficiently close to, but inside, the unit circle. To check the normalization, fix $y = e^{\mathrm{i}q}$ and take the x residue at the interior root $\rho = e^{-\gamma}$. Since

$$D_x(\rho,y) = \frac{S(1-\rho^2)}{2\rho^2}, \qquad \frac{\sqrt{p}}{D_x(\rho,y)} = \sqrt{B}\frac{\rho}{\sinh\gamma},$$

and subtraction of two dispersion equations gives

$$\mathscr{A}(\rho_i,\rho_j)\mathscr{A}(y_i,y_j) = -B\frac{(y_i-y_j)^2\rho_i\rho_j}{y_iy_j(1-\rho_i\rho_j)^2} = B\left[\frac{\sin((q_i-q_j)/2)}{\sinh((\gamma_i+\gamma_j)/2)}\right]^2 =: P_{ij}, \tag{1.11}$$

the residue factors supply $B^{N/2}$ and the pair factors supply $B^{N(N-1)/2}$. This recovers precisely $B^{N^2/2}$. The remaining factors in (1.10) give the stated Fourier and spatial decay factors. Each separation can be reflected independently at the fixed coupling ratio.

The exact full-site multiplier in the double contour is

$$\frac{(1+X)(1+Y)}{XY(1-X)(1-Y)} = \frac{1}{XY} + \frac{2(X^{-1}+Y^{-1})}{(1-X)(1-Y)}. \tag{1.12}$$

It is the product of the two sums $\sum_{m\in\mathbb{Z}} X^{|m|-1}$ and $\sum_{l\in\mathbb{Z}} Y^{|l|-1}$. It counts the origin once, each reflected axis correctly, and all four open quadrants. If C_N is the full-site coefficient and T_N the contour with the second numerator but without its factor two, then $C_N = f_N(0,0) + 2T_N$. The onsite sums are one in the HT phase and $1-M^2$ after the LT subtraction. This decomposition retains the axes, the origin, and the open quadrants within the same coefficient.

Summing the Fourier variable and the geometric spatial series also gives the complete angular representation

$$C_N = \frac{1}{N!}\int_{\sum q_i = 0 \bmod 2\pi} G_N(q)\frac{1+Z}{1-Z}\prod_{i=1}^{N-1}\frac{\mathrm{d}q_i}{2\pi}, \quad G_N = B^{N/2}\prod_i \frac{1}{\sinh\gamma_i}\prod_{i<j} P_{ij}, \quad Z = \prod_i e^{-\gamma_i}. \tag{1.13}$$

There are exactly $N-1$ independent normalized angular measures, with $q_N = -\sum_{i<N} q_i$ modulo 2π.

1.4 A holomorphic full sum on an independent spectral domain

Define

$$\mathcal{S}_\lambda = \{z : \exists u, v \in [-1,1],\ A(z) = S(z)u + T(z)v\}. \tag{1.14}$$

It is exactly the set where (1.9) has a zero on the product of two unit circles. Section 2 proves that the components of $\mathbb{C}\setminus\mathcal{S}_\lambda$ containing $(0, z_c)$ and (z_c, ∞) are exactly (1.5), and that both are simply connected.

We first establish local normal convergence of the particle series on the spectral complement. The physical prefactors are attached on the two specified components below. The later boundary estimates will determine whether those physical sums can extend further. On a compact subset of the spectral complement the torus polynomial has a nonzero minimum modulus. Move both radii a small common amount inside one, within a product annulus where this gap persists. The resulting contours can be held fixed on a parameter neighborhood. Their Schur entries are bounded. The Schur Pfaffian identity, followed by Hadamard's determinant inequality, bounds the product of the two Pfaffians by $C^N N^{(N+1)/2}$, with the harmless extra odd dimension obtained by adjoining a constant ghost entry. After the $1/N!$ normalization and the bounded one-body and full-site factors this is summable in N. It is locally normally summable together with every fixed parameter derivative by Cauchy's inequality on a slightly larger neighborhood.

On a physical real compact set the same contours in (1.10) additionally give $r_x^{N|m|} r_y^{N|l|}$ with $r_x, r_y < 1$. This proves absolute summability in the lattice sites and particles and justifies their interchange. Thus (1.13) and the double contour define the same physical germs. Absolute summability supplies the full-site identification and the interchange used to obtain it.

Write $F_N = p^{-N/2} C_N$ for the unscaled double contour, whose integrand uses (1.9) and (1.12). The complete sums are

$$\mathcal{X}_H = (1-p^2)^{1/4} \sum_{N\geq 1,\ N \text{ odd}} p^{(N-1)/2} F_N, \tag{1.15}$$

$$\mathcal{X}_L = (1-p^{-2})^{1/4} \sum_{N\geq 2,\ N \text{ even}} p^{N/2} F_N. \tag{1.16}$$

The powers inside both sums are integers. The identity $A^2 = (1+S^2)(1+T^2)$ implies that every $p^2 = 1$ point lies in $\mathcal{S}_\lambda$. Also, p never vanishes in $\Omega_L \subset \{\Re z > 0\}$; in Ω_H its only zero is the double zero at the origin. Hence the prefactors in these formulas have unique holomorphic branches on the specified simply connected domains, fixed by the real physical values. Formula (1.15) removes the apparent HT $1/\sqrt{p}$ singularity at zero; $\mathcal{X}_H(0) = 1$. In particular, the one-particle term is

$$\mathcal{X}_H^{(1)} = \frac{(1-p^2)^{1/4}}{A-S-T},$$

which gives $1 + 2\beta(J_x + J_y) + O(\beta^2)$ at the origin.

For the angular formulas the spectral root is globally specified by

$$\rho = W - \sqrt{W^2-1}, \qquad |\rho| < 1, \qquad W = (A - T\cos q)/S.$$

The square root has asymptotic W at infinity. For z in the spectral complement, W avoids $[-1,1]$ for every real q, so the inside root is unique. At $S = 0$ its root equation $S\rho^2 - 2(A - T\cos q)\rho + S = 0$ gives its removable value zero. A local logarithm $\gamma = -\log\rho$ is used only where nonzero and needed. No unsupported global logarithm or change of transfer vacuum is imposed by periodic complex coupling shifts.

2 Exact geometry on every fixed positive coupling slice

This section determines where the contour denominator can meet the integration torus. The sign factorization gives the exact spectral set; two components of its complement give the lens and epigraph. For unequal couplings, the final steps isolate the regular arcs on which the analytic estimates will be performed and show that the omitted points are locally discrete.

Fix $\lambda > 0$, and use the one-complex-variable coordinate

$$z = 2\beta J_x = x + \mathrm{i}y, \qquad a(z) = \sinh z, \quad b(z) = \sinh(\lambda z),$$
$$A(z) = \cosh z \cosh(\lambda z), \qquad p(z) = a(z)b(z).$$

Here $K_y = \lambda K_x$ throughout, and b/a varies meromorphically with z.

2.1 The spectral set

Definition 2.1. For $(u,v) \in [-1,1]^2$, let

$$f_\lambda(z;u,v) = A(z) - a(z)u - b(z)v.$$

The spectral set is

$$\mathcal{S}_\lambda = \{z \in \mathbb{C} : \ f_\lambda(z;u,v) = 0 \text{ for some } (u,v) \in [-1,1]^2\}. \tag{2.1}$$

Equivalently, it is the set where the elementary spectral polynomial

$$A(z) - \frac{a(z)}{2}(X + X^{-1}) - \frac{b(z)}{2}(Y + Y^{-1})$$

has a zero with $|X| = |Y| = 1$.

Restricting (2.1) to cosine grids gives the anisotropic Nickel equations [4, eq. (54)]. The grid closure lemma below identifies their accumulation set. The boundary singularities of the physical sum are established in Sections 3–5.

Put

$$D(z) = \Im(\overline{a(z)}b(z)), \qquad U(z) = \frac{\Im(\overline{A(z)}b(z))}{D(z)}, \qquad V(z) = \frac{\Im(\overline{a(z)}A(z))}{D(z)}. \tag{2.2}$$

When $D \neq 0$, the last two quantities are the unique real solution of $A = aU + bV$. Thus membership means $|U|, |V| \leq 1$, and strict inequalities give an interior point of $\mathcal{S}_\lambda$. At $D = 0$ and $a \neq 0$, membership instead means

$$b/a \in \mathbb{R}, \qquad A/a \in \mathbb{R}, \qquad |A/a| \leq 1 + |b/a|.$$

If $a = 0$, the analogous condition with b applies. Simultaneous zeros of a, b never belong to $\mathcal{S}_\lambda$, since then $A = \pm 1$. These conditions retain the unsquared equation and its sign information.

Lemma 2.2 (Closure of both parity grid families)**.** *For each parity separately, the closure of the union of zeros*

$$f_\lambda(z; \cos(2\pi j/n), \cos(2\pi k/n)) = 0$$

is exactly $\mathcal{S}_\lambda$. Odd prime orders $n = q$, and even orders $n = 2q$ with q an odd prime, already suffice. The assertion remains true after deleting grid pairs in any convention that leaves the grids dense in $[-1, 1]^2$, including deletion of the isotropic exceptional pairs $j = k = 0$, and $j + k = n/2$ when n is even.

Proof. The set $\mathcal{S}_\lambda$ is closed: for a convergent sequence of its points, extract a convergent subsequence of representing pairs from the compact square and pass to the limit in f_λ. This proves one inclusion.

For the reverse inclusion, fix $z_0 \in \mathcal{S}_\lambda$ with representing pair (u, v), and choose allowed grid pairs $(u_m, v_m) \to (u, v)$. The functions $f_\lambda(\cdot; u_m, v_m)$ converge uniformly on compact sets to $f_\lambda(\cdot; u, v)$. The latter is not identically zero, since its value at $z = 0$ is one. On a sufficiently small circle around z_0 it has no zeros, and Rouche's theorem supplies a zero of every sufficiently close approximating function inside that circle. A sequence of shrinking circles gives zeros converging to z_0. This argument also covers multiple roots, endpoints, $D = 0$, and zeros of either hyperbolic sine.

The indicated prime grids are dense because their angular mesh tends to zero. Removing the displayed exceptional pairs does not affect density: when a nearest pair is deleted, changing one index by one step, or a bounded number of steps, avoids the exceptional equalities while the mesh error still tends to zero. □

The deletions in Lemma 2.2 are the isotropic convention in [4, eq. (52)]. The physical parity restrictions come from the correlation expansions.

Lemma 2.3. *The set $\mathcal{S}_\lambda$ has no isolated points.*

Proof. At a point $z_0 \in \mathcal{S}_\lambda$, the coefficients $a(z_0), b(z_0)$ do not both vanish. Perturb a representing pair into the open square so that $f_\lambda(z_0; u_m, v_m) \neq 0$, which is possible because that affine function of the pair is not constant. The preceding Rouche argument produces distinct nearby roots tending to z_0. □

2.2 A closed real-analytic inequality, including every degeneracy

The two real functions below encode the spectral membership test without dividing by the real-linear determinant. The proof first handles opposite and equal signs, then treats endpoint and collinear cases.

Define

$$F_1(x,y) = \cosh^2(\lambda x)\cos^2 y - \cosh^2 x \sinh^2(\lambda x), \tag{2.3}$$

$$F_2(x,y) = \cosh^2 x \cos^2(\lambda y) - \sinh^2 x \cosh^2(\lambda x). \tag{2.4}$$

Theorem 2.4 (Exact fixed-slice geometry)**.** *For every $\lambda > 0$,*

$$\boxed{\mathcal{S}_\lambda = \{x + \mathrm{i}y : \ F_1(x,y)F_2(x,y) \le 0\}.} \tag{2.5}$$

Proof. Reflection in the imaginary axis preserves the set and the two real functions, so it is enough to consider $x \ge 0$. At $x = 0$, a, b are purely imaginary and A is real. Membership is exactly

$$A = \cos y \cos(\lambda y) = 0,$$

which is precisely $F_1F_2 = 0$ there.

Suppose now $x > 0$, and abbreviate

$$h = \sinh x, \quad c = \cosh x, \quad H = \sinh(\lambda x), \quad C = \cosh(\lambda x),$$
$$t = \cos y, \quad q = \sin y, \quad T = \cos(\lambda y), \quad Q = \sin(\lambda y).$$

Write $N_U = \Im(\overline{A}b)$, $N_V = \Im(\overline{a}A)$. Expansion gives the real identities

$$D = hCtQ - cHqT, \qquad N_U = ctTQ - hHCq, \qquad N_V = hcHQ - CtqT,$$

and the exact factorizations

$$N_U^2 - D^2 = (c^2T^2 - h^2C^2)(t^2Q^2 - H^2q^2), \tag{2.6}$$

$$N_V^2 - D^2 = (c^2H^2 - C^2t^2)(h^2Q^2 - q^2T^2). \tag{2.7}$$

For clarity set

$$a_0 = h^2, \ b_0 = H^2, \ A_0 = 1 + a_0, \ B_0 = 1 + b_0, \quad X = t^2, \ Y = T^2, \quad X_0 = \frac{A_0 b_0}{B_0}, \quad Y_0 = \frac{a_0 B_0}{A_0}.$$

Then $F_1 = B_0(X - X_0)$, $F_2 = A_0(Y - Y_0)$, and

$$N_U^2 - D^2 = F_2 K, \qquad K = (B_0 - Y)X - b_0, \tag{2.8}$$

$$N_V^2 - D^2 = -F_1 L, \qquad L = a_0 - (A_0 - X)Y. \tag{2.9}$$

Both K, L vanish at (X_0, Y_0), increase with X, and decrease with Y. Thus $F_1 > 0, F_2 < 0$ implies $K, L > 0$, whereas $F_1 < 0, F_2 > 0$ implies $K, L < 0$. In either case the right-hand sides in (2.8)–(2.9) are negative. In particular $D \neq 0$, since otherwise a real square would be negative, and $|U|, |V| < 1$. This proves membership for strict opposite signs directly, including the needed nonvanishing of D.

If $D \neq 0$, $0 < Y < 1$, and both F's are positive, the two membership inequalities would require

$$X \leq \frac{b_0}{B_0 - Y}, \qquad X \geq A_0 - \frac{a_0}{Y}.$$

They are inconsistent because $Y > Y_0$ and

$$\frac{b_0}{B_0 - Y} - \left(A_0 - \frac{a_0}{Y}\right) = \frac{A_0(Y-1)(Y-Y_0)}{Y(B_0 - Y)} < 0. \tag{2.10}$$

If both F's are negative, the required inequalities reverse; now $Y < Y_0$ makes (2.10) positive, again inconsistent. At $Y = 0$, the factors directly give membership exactly when $X \geq X_0$. At $Y = 1$, they give membership exactly when $Y_0 \leq 1$ and $X \leq X_0$. These conditions agree with $F_1 F_2 \leq 0$; if $Y_0 > 1$, then also $X_0 > 1$, so the opposite sign possibility at $Y = 1$ cannot occur.

It remains to handle zeros and collinearity. Every point with $F_1 = 0$ belongs to the spectral set, including those with $D = 0$. Indeed, the two endpoint dispersion root pairs are

$$r_+(z) = e^{\lambda z} \tanh(z/2), \qquad \frac{r_+ + r_+^{-1}}{2} = \frac{A - b}{a}, \tag{2.11}$$

$$r_-(z) = e^{-\lambda z} \tanh(z/2), \qquad \frac{r_- + r_-^{-1}}{2} = \frac{A + b}{a}. \tag{2.12}$$

Here $a \neq 0$, and $\tanh(z/2)$ has neither zero nor pole because $x > 0$. The exact modulus identity

$$|\tanh(z/2)|^2 = \frac{\cosh x - \cos y}{\cosh x + \cos y}$$

shows that $|r_+| = 1$ exactly when $\cos y = \cosh x \tanh(\lambda x)$, and $|r_-| = 1$ exactly when the right side has a minus sign. These are the two possibilities in $F_1 = 0$. A unit-modulus root has $(r + r^{-1})/2 \in [-1, 1]$, giving respectively the representing endpoint $V = 1$ or $V = -1$. Exchanging the two axes proves the same assertion for $F_2 = 0$.

Conversely suppose $D = 0$ at a candidate point. Then $N_U = N_V = 0$. If both F's were nonzero, (2.8)–(2.9) would give $K = L = 0$. These force $Y > 0$, and (2.10) forces $Y = Y_0$ or $Y = 1$. The first contradicts $F_2 \neq 0$. In the second case $X = 1$ too. The coefficients are then real up to their signs, and candidate membership requires

$$\cosh x \cosh(\lambda x) \leq \sinh x + \sinh(\lambda x).$$

The square of the left side minus the square of the right side is $(\sinh x \sinh(\lambda x) - 1)^2$. Equality is therefore forced, and $F_1 = F_2 = 0$, again a contradiction. Finally, at $F_1 = 0$ identity (2.8) becomes $N_U^2 - D^2 = -(b_0/B_0)F_2^2$; at $F_2 = 0$, (2.9) becomes $N_V^2 - D^2 = -(a_0/A_0)F_1^2$. Thus a collinear candidate with one zero has both zeros. The sign, endpoint, and collinearity cases together prove (2.5). □

2.3 The two exact physical spectral components

The sign criterion now identifies the components selected by the two positive real temperature intervals. The inverse profiles translate the two sign inequalities into explicit boundary curves.

Let $z_c > 0$ be the unique solution

$$\sinh z_c \sinh(\lambda z_c) = 1. \tag{2.13}$$

Define strictly increasing functions on $[0,\infty)$ by

$$P_\lambda(t) = \cosh t \tanh(\lambda t), \qquad Q_\lambda(t) = \cosh(\lambda t)\tanh t. \tag{2.14}$$

Both start at zero, tend to infinity, and satisfy $P_\lambda(z_c) = Q_\lambda(z_c) = 1$. Their real inverse functions are unambiguous. Put

$$Y_\lambda = \frac{\pi}{2\max(1,\lambda)}, \tag{2.15}$$

$$H_\lambda(y) = \min\{P_\lambda^{-1}(\cos y), Q_\lambda^{-1}(\cos(\lambda y))\}, \qquad |y| < Y_\lambda, \tag{2.16}$$

$$L_\lambda(y) = \max\{P_\lambda^{-1}(|\cos y|), Q_\lambda^{-1}(|\cos(\lambda y)|)\}, \qquad y \in \mathbb{R}. \tag{2.17}$$

Theorem 2.5 (Physical spectral components)**.** *The component of $\mathbb{C} \setminus \mathcal{S}_\lambda$ containing the real high-temperature interval $(0, z_c)$, and the component containing the real low-temperature interval (z_c, ∞), are respectively*

$$\Omega_H = \{x + \mathrm{i}y : \ |y| < Y_\lambda, \ |x| < H_\lambda(y)\}, \tag{2.18}$$

$$\Omega_L = \{x + \mathrm{i}y : \ x > L_\lambda(y)\}. \tag{2.19}$$

Both are simply connected. Their boundaries are the complete lens boundary and the complete graph $x = L_\lambda(y)$, respectively. Moreover $0 \le L_\lambda(y) \le z_c$, and $\mathcal{S}_\lambda \subset \{|\Re z| \le z_c\}$.

Proof. At the origin F_1, F_2 are both positive. Their signs cannot change in its complement component. Since neither cosine can then vanish, both retain their positive signs, giving $|y| < Y_\lambda$. The two positive inequalities are exactly $|x| < H_\lambda(y)$. Conversely this entire lens is connected: decrease $|x|$ to zero at fixed y, then decrease $|y|$ to zero. Both operations preserve the inequalities. In fact the lens is star shaped about zero and hence simply connected.

At real $z > z_c$, both F's are negative. For $x > 0$, their simultaneous negativity is exactly $x > L_\lambda(y)$. This epigraph is connected because $L_\lambda \le z_c$, and a component with both signs negative cannot cross $x = 0$, where both are nonnegative. Thus it is precisely the LT component. Its continuous boundary graph is mapped to a straight line by the homeomorphism $(x, y) \mapsto (x - L_\lambda(y), y)$, proving simple connectivity. Finally if $|x| > z_c$, both $P_\lambda(|x|), Q_\lambda(|x|)$ exceed one, so both F's are negative and the point is outside $\mathcal{S}_\lambda$. □

The boundary functions are explicitly defined before any analyticity or singularity assertion for a particle sum. In particular the set $\mathcal{S}_\lambda$ itself generally has interior, whereas the boundaries in Theorem 2.5 are one-dimensional barriers.

Proposition 2.6 (Strict anisotropic opening)**.** *For every $\lambda \ne 1$, $\mathcal{S}_\lambda$ has nonempty interior arbitrarily close to z_c. The horizontal width between the two local endpoint curves near $z_c + \mathrm{i}y$ is*

$$\frac{1}{2}|\lambda \coth(\lambda z_c) - \coth z_c| y^2 + O(y^4). \tag{2.20}$$

Proof. For small y, the two zero curves are

$$x_1(y) = P_\lambda^{-1}(\cos y) = z_c - \frac{y^2}{2P_\lambda'(z_c)} + O(y^4),$$

$$x_2(y) = Q_\lambda^{-1}(\cos(\lambda y)) = z_c - \frac{\lambda^2 y^2}{2Q_\lambda'(z_c)} + O(y^4).$$

Using $\sinh z_c \sinh(\lambda z_c) = 1$, subtraction of the two displayed coefficients gives (2.20). The coefficient is nonzero for $\lambda \neq 1$: the function $t \mapsto t \coth t$ is strictly increasing for $t > 0$, since $\sinh t \cosh t - t > 0$. Strictly between the two distinct curves the F's have opposite signs, so Theorem 2.4 gives an open part of the spectral set. □

For a concrete exact check, at $\lambda = 2$ take

$$a_0 = \frac{1 + \mathrm{i}\sqrt{7}}{4}, \qquad c_0 = \sqrt{1 + a_0^2}, \quad \Re c_0 > 0, \qquad z_0 = \operatorname{arsinh} a_0$$

on the corresponding local branch. Then $b = 2a_0c_0$, $A = c_0(1 + 2a_0^2) = b/2$, so $U = 0, V = 1/2$. Also $D = \Im c_0 \neq 0$ and $|ab| = 2^{-1/4} \neq 1$. Thus the full spectral set differs from $|ab| = 1$. This explicit interior point illustrates why the natural-boundary statement concerns the boundaries of the physical complementary components.

2.4 Zeros, collinearity and every exceptional boundary point

To use regular edge coordinates densely, we must account for their exceptional points. We treat elementary zeros first, then collinearity, and finally the regular-arc density argument.

The imaginary-axis intersection has the exact simple description

$$\mathcal{S}_\lambda \cap \mathrm{i}\mathbb{R} = \{\mathrm{i}y : \cos y \cos(\lambda y) = 0\}. \tag{2.21}$$

It is locally finite. Both cosines vanish simultaneously if and only if λ is a ratio of two positive odd integers. In reduced form $\lambda = r/s$ with both r, s odd, these simultaneous points have $y = (2m + 1)\pi s/2$.

A zero of a has $z = \mathrm{i}\pi m$ and belongs to $\mathcal{S}_\lambda$ exactly when $\cos(\lambda\pi m) = 0$; then $b = \pm\mathrm{i}$, $A = 0$, and its representing pairs have $V = 0$. The corresponding statement for $b = 0$ follows by axis exchange. Simultaneous sine zeros are outside the spectral set. Inside Ω_H, the only sine zero is the common zero at $z = 0$; inside Ω_L neither sine vanishes. In addition,

$$\bigcup_{\varepsilon_1, \varepsilon_2 \in \{1, -1\}} \{A = \varepsilon_1 a + \varepsilon_2 b\} = \{p^2 = 1\} \subset \mathcal{S}_\lambda, \tag{2.22}$$

because

$$A^2 - (\varepsilon_1 a + \varepsilon_2 b)^2 = (p - \varepsilon_1\varepsilon_2)^2.$$

Every such set is discrete on the fixed nonconstant slice. Hence the phase prefactors have no zero or branch point in either physical component, except for the removable HT normalization issue at the origin treated by its integer-power formula. On Ω_L, even that issue is absent. The boundary of Ω_L reaches the imaginary axis only at simultaneous cosine zeros.

Lemma 2.7 (Collinear candidates)**.** *For $\Re z \neq 0$,*

$$\{z \in \mathcal{S}_\lambda : D(z) = 0\} = \{z : D(z) = 0,\ |a(z)b(z)| = 1\}.$$

For each $\lambda > 0$, $\lambda \neq 1$, this set, together with the collinear candidates on the imaginary axis, is locally discrete.

Proof. Off the imaginary axis, $a, b \neq 0$. Write $b = ra$, $r \in \mathbb{R} \setminus \{0\}$. A candidate also has $A = ta$, $t \in \mathbb{R}$, and

$$t^2 = 1 + r^2 + r^2a^2 + a^{-2}.$$

If $a = Re^{\mathrm{i}\theta}$, reality of this expression implies $(r^2R^2 - R^{-2})\sin(2\theta) = 0$. Unless a is real or purely imaginary, this gives $|ab| = 1$. If a, b are real, the inequality $|A| \leq |a| + |b|$, together with

$$|A|^2 - (|a| + |b|)^2 = (|ab| - 1)^2,$$

gives the same conclusion. If both are purely imaginary off the imaginary axis, then both cosines vanish, $|a|, |b| > 1$, and A is nonzero real; it cannot be a real multiple of imaginary a. Conversely, if $D = 0$ and $|ab| = 1$, then

$$(A/a)^2 = 1 + r^2 + 2|r|\cos(2\theta) \in [(1 - |r|)^2, (1 + |r|)^2].$$

Thus A/a is real with the required segment bound, proving the set equality.

By symmetry it is enough to prove local discreteness for $x > 0$. For each y with $|\sin y \sin(\lambda y)| < 1$, the equation

$$(\sinh^2 x + \sin^2 y)(\sinh^2(\lambda x) + \sin^2(\lambda y)) = 1 \tag{2.23}$$

has a unique solution $x = X(y) > 0$. The left side strictly increases with x, starts below one, and tends to infinity. Its positive x derivative proves that X is real analytic. At a simultaneous cosine zero it tends continuously to zero.

If there are no simultaneous cosine zeros, X is analytic on all of $\mathbb{R}$. The analytic function $D(X(y) + \mathrm{i}y)$ is not identically zero, because its derivative at $y = 0$ is

$$a(z_c)b'(z_c) - a'(z_c)b(z_c) = \lambda \coth(\lambda z_c) - \coth z_c \neq 0.$$

Its zeros are therefore locally discrete. If simultaneous cosine zeros occur, write $\lambda = r/s$ in reduced odd/odd form. The intervals between $(2m + 1)\pi s/2$ are translates by πs of one another, and each contains a translate of $y = 0$. Both X and the displayed restriction of D have that period, so the restriction is not identically zero on any interval.

There is no accumulation of these zeros at the interval endpoints. At a simultaneous zero, write $z = z_0 + \zeta$, $\zeta = x + \mathrm{i}\delta$. For signs $\sigma, \tau \in \{1, -1\}$,

$$a = \mathrm{i}\sigma \cosh \zeta, \qquad b = \mathrm{i}\tau \cosh(\lambda\zeta), \qquad D = \sigma\tau(\lambda^2 - 1)x\delta + O(|\zeta|^4).$$

Equation (2.23) gives $X(y) = |\delta| + O(|\delta|^3)$, so $D \neq 0$ for all sufficiently small nonzero δ. At any other imaginary-axis candidate, $|ab| < 1$, and continuity excludes approach by off-axis collinear candidates with $|ab| = 1$. This proves local discreteness everywhere. □

Definition 2.8. A regular edge point has $D \neq 0$, exactly one of U, V equal to 1 or -1, the other strictly in $(-1, 1)$, and nonzero derivative of the corresponding meromorphic edge map

$$h_\varepsilon(z) = \frac{A(z) - \varepsilon a(z)}{b(z)} \quad (U = \varepsilon), \qquad g_\varepsilon(z) = \frac{A(z) - \varepsilon b(z)}{a(z)} \quad (V = \varepsilon).$$

Proposition 2.9 (Dense regular boundary arcs)**.** *For every fixed $\lambda > 0$, $\lambda \neq 1$, regular edge points are dense in the boundary of every component of $\mathbb{C} \setminus \mathcal{S}_\lambda$, in particular in both boundaries in Theorem 2.5. The excluded set is locally discrete.*

Proof. The four edge maps are nonconstant meromorphic functions: each has a pole at $z = 0$. Their critical points and poles are locally discrete. Corner points are the discrete zeros in (2.22); collinear candidates are locally discrete by Lemma 2.7. Let E be their union, restricted to $\mathcal{S}_\lambda$.

At a boundary point outside E, strict inequalities cannot hold for both coordinates, since they would give an interior point. Exactly one is an endpoint, and the corresponding holomorphic edge map has nonzero derivative. Its imaginary part is a nondegenerate local defining function for that edge. The unique real-coordinate solution in (2.2) then shows that $\mathcal{S}_\lambda$ is locally one closed side of an analytic arc and its complement is the other side.

If Ω is any complement component, then $\partial\Omega \subset \partial\mathcal{S}_\lambda$. It has no isolated point: if z_0 were isolated in $\partial\Omega$, a sufficiently small punctured disk, being connected and containing points of Ω, would lie wholly in Ω. Then z_0 would be an isolated point of $\mathcal{S}_\lambda$, contrary to Lemma 2.3. Removing the locally discrete set E from a boundary without isolated points leaves a dense subset. □

At the right side of the physical HT lens, both cosines in (2.16) are positive. Equations (2.11)–(2.12), or their axis-swapped versions, show that every regular edge there is a positive endpoint. Reflection $z \mapsto -\overline{z}$ preserves the lens and sends the representing pair (U, V) to $(-U, -V)$, so every regular left edge is a negative endpoint. Thus negative endpoints belong to the physical HT boundary for every $\lambda \neq 1$. They require the separate odd-sum argument of Sections 4–5. On the LT boundary, an $F_1 = 0$ regular edge has $V = \operatorname{sign}(\cos y)$, and an $F_2 = 0$ regular edge has $U = \operatorname{sign}(\cos(\lambda y))$. These alternatives cover all its regular pieces.

2.5 The canonical root and the precise resonance arithmetic

Lemma 2.10 (Canonical spectral sheet)**.** *For $z \notin \mathcal{S}_\lambda$ and real q, the equation*

$$a\rho^2 - 2(A - b\cos q)\rho + a = 0$$

has a unique root with $|\rho| < 1$, interpreted as $\rho = 0$ if $a = 0$. It is holomorphic in z, locally jointly holomorphic with the complex angle, and has a common modulus gap below one on every compact subset of $\mathbb{C} \setminus \mathcal{S}_\lambda$ and the real angle circle. On the positive real HT and LT intervals it is the positive physical decaying root.

Proof. When $a \neq 0$, put $W = (A - b\cos q)/a$. Outside the spectral set, $W \notin [-1, 1]$. The root

$$\rho(W) = W - \sqrt{W^2 - 1}, \qquad \sqrt{W^2 - 1} \sim W \quad (W \to \infty),$$

is the unique interior root on $\mathbb{C} \setminus [-1, 1]$; the other root is its reciprocal. At $a = 0$, the coefficient $A - b\cos q \neq 0$ by the spectral-set definition, so the implicit-function theorem extends the root through $\rho = 0$. Compactness supplies the uniform gap and a thin complex angle neighborhood. For positive real $z \neq z_c$, $A - a - b > 0$, since

$$(A - a - b)(A + a + b) = (ab - 1)^2,$$

and hence $W(q) > 1$ for all real q, proving the physical match. □

On the physical LT component the root is nonzero, and simple connectivity fixes the algebraic prefactors from their real values. On the HT component Section 1 gives the regular normalization at the sole root zero, the origin.

At a regular edge with one cosine equal to $+1$ and the other equal to $\cos(2\pi k/q)$, where q is an odd prime and $1 \leq k \leq (q-1)/2$, uniqueness in (2.2) implies that the least grid order is

q and the least even order is $2q$. These values are dense on every regular edge arc because its nonramified edge coordinate maps the arc to a real interval. At a -1 endpoint the grid order must be even, and the least such order is $2q$. The integral estimates below use these first resonances and establish lower-order regularity and noncancellation.

An explicit family of grid roots approaches a negative endpoint from the interior of $\mathcal{S}_\lambda$. At a regular $U = -1$ point with $V = \cos(2\pi k/q)$, take odd $n = q(2m+1)$ and set

$$u_n = -\cos(\pi/n), \qquad v_n = V.$$

The simple root of the candidate equation has

$$z_n = z_0 + \frac{a(z_0)}{\partial_z f_\lambda(z_0; -1, V)} \frac{\pi^2}{2n^2} + O(n^{-4}).$$

It lies in the interior of $\mathcal{S}_\lambda$ for large n, and its least grid order is n, since $(n-1)/2$ is coprime to n. The derivative in the denominator is nonzero by nonramification of the edge map.

2.6 Isotropic covering and ratio symmetries

At $\lambda = 1$, $a = b = s = \sinh z$ and $A = 1 + s^2$, so

$$A/a = s + s^{-1} = u + v \in [-2, 2].$$

The preimage of $[-2, 2]$ under the Joukowski map $s + s^{-1}$ is exactly $|s| = 1$. Thus

$$\mathcal{S}_1 = \{z : |\sinh z| = 1\}. \tag{2.24}$$

Equivalently, $F_1 = F_2$, and their common zero is $\cos^2 y = \sinh^2 x$. Since $P_1 = Q_1 = \sinh$, the two physical components become

$$\Omega_H(1) = \{|y| < \pi/2,\ |x| < \operatorname{arsinh}(\cos y)\}, \qquad \Omega_L(1) = \{x > \operatorname{arsinh}(|\cos y|)\}.$$

The map $s = \sinh z$ maps the first biholomorphically onto the unit disk, with the inverse-sinh branch real at the origin. It maps the second as the universal cover of $|s| > 1$, with deck translations $z \mapsto z + 2\pi\mathrm{i}$. To see the latter directly, put $w = e^z$. For $|s| > 1$, the equation $w - w^{-1} = 2s$ has exactly one root with $|w| > 1$, depending holomorphically on s. A unit-modulus root would instead give $s \in \mathrm{i}[-1, 1]$. The exterior w domain is a continuous radial epigraph; exponentiation from $\Omega_L(1)$ is its universal cover. This proves the covering assertion.

At the boundary points with $\cosh z = 0$, the inverse-sinh coordinate is branched. Elsewhere on the boundary it is locally biholomorphic. The dense isotropic obstruction lifts at the nonramified points and includes the isolated branch points by density.

Finally $\mathcal{S}_\lambda$ is invariant under conjugation and negation, and axis exchange gives

$$\mathcal{S}_{1/\lambda} = \lambda \mathcal{S}_\lambda.$$

If $\lambda = r/s$ is rational in lowest positive terms, $\mathrm{i}\pi s$ is an imaginary period of the spectral set: the two representing cosines change by signs. For irrational λ, there is no nonzero imaginary period. Indeed (2.21) is the union of two arithmetic progressions with spacings π and π/λ. A translate of the first progression can intersect the second in at most one point unless their spacing ratio is rational. A period of the union must therefore be a period of each progression, forcing it to belong to both $\pi\mathbb{Z}$ and $(\pi/\lambda)\mathbb{Z}$, hence to be zero.

For each $\lambda \neq 1$, the regular arcs are dense in the boundary. The exceptional points include the odd-over-odd simultaneous cosine zeros, elementary zeros, corners and critical points of the meromorphic edge coordinates.

3 Regular spectral edges: leading amplitudes and tail estimates

At a regular boundary point exactly one of the two real spectral coordinates is $+1$ or -1, the other lies strictly between them, and the corresponding edge map is a nonramified holomorphic local coordinate. Interchange the transfer axes when necessary. First take the endpoint $q_0 = 0$ and write

$$W(q;u) = b + u + B(u)(1-\cos q), \qquad b = \cos\phi, \quad 0 < \phi < \pi, \quad \Im B(0) > 0. \tag{3.1}$$

Here $u = W(0) - b$ vanishes at the selected point. The opposite sign of $\Im B$ is obtained by complex conjugation. On the edge, $W(q)$ is in the upper half-plane except at $q = 0$. Its canonical root $\rho(q) = e^{-\gamma(q)}$ lies in the lower half-disk, with $\rho(0) = e^{-\mathrm{i}\phi}$. Compactness gives

$$\Im\rho(q) \le -\delta < 0, \qquad |\rho(q)| \le 1 \quad (q \in \mathbb{R}). \tag{3.2}$$

There is no real-angle point with $W = \pm 1$. The root, a local γ, and all one-body and canceled-pair denominators are therefore holomorphic on a common fixed complex parameter and angular neighborhood. On the physical-domain approach $u = \mathrm{i}\varepsilon$, $\varepsilon > 0$,

$$\Re\gamma(q;\mathrm{i}\varepsilon) \ge c\big(\varepsilon + \operatorname{dist}(q,0)^2\big). \tag{3.3}$$

Indeed $\Im W = \varepsilon + \Im B(\mathrm{i}\varepsilon)(1-\cos q)$ is bounded below by that expression, while $\Im W = \sinh(\Re\gamma)\sin(\Im\gamma)$ on the compact chosen branch. The same local side belongs to the physical component bordering the regular edge, by the geometry of Section 2.

3.1 Schur contraction and the form-factor numerator

The matching identity combines strict contraction of the spectral Schur factors with boundedness of the complete canceled pairs. The elementary identity

$$|1 - vw|^2 - |v - w|^2 = (1 - |v|^2)(1 - |w|^2) + 4\Im v\,\Im w \tag{3.4}$$

and (3.2) give a constant $\rho_0 < 1$ such that $|\mathscr{A}(\rho_i, \rho_j)| \le \rho_0$ for every pair of real angles. By continuity a slightly larger constant still below one works on a fixed complex parameter neighborhood. The bound on this neighborhood follows by continuity at the boundary and applies also where $|\rho_i| > 1$. The canceled pairs P_{ij} are analytic and bounded there.

The canceled factors can exceed one. For example, on (3.1),

$$P(q,-q) = \frac{B\sin^2 q}{[b + B(1-\cos q)]^2 - 1}.$$

When $\Im B$ is small and the real part of the bracket is close to one, this can be arbitrarily large. The matching argument uses strict contraction for the uncanceled ρ-Schur factors.

Lemma 3.1 (Matched-Pfaffian numerator estimate)**.** *For every fixed integer $r \ge 0$, on a common parameter disk,*

$$|\partial_u^r G_N(q;u)| \le C_r \exp\{-\kappa N^2 + CN\log(N+1)\}$$

uniformly in the real angles and all positive N. The constants may depend on the edge but not on N.

Proof. For even N, expand only the Schur Pfaffian in the $y_i = e^{\mathrm{i}q_i}$ variables. Each of its $(N-1)!!$ matchings contains $N/2$ edges. Combine each matched y factor with its corresponding ρ factor using (1.11); bound that complete pair by a constant C. The other $N(N-2)/2$ factors are bounded by ρ_0. Thus

$$\left|\prod_{i<j} P_{ij}\right| \le (N-1)!!\, C^{N/2} \rho_0^{N(N-2)/2}.$$

For odd N, adjoin a ghost variable -1 to the y Pfaffian. Its Schur entry with every real variable is the removable constant one. There are $N!!$ matchings, with $(N-1)/2$ real matched edges and one ghost edge, leaving $(N-1)^2/2$ uncanceled ρ factors. This gives

$$\left|\prod_{i<j} P_{ij}\right| \le N!!\, C^{(N-1)/2} \rho_0^{(N-1)^2/2}.$$

The identities initially hold away from reciprocal y pairs and extend by the analytic canceled formula, which involves no division by a vanishing ρ factor. One-body factors, including $B^{N/2}$, cost C^N. The unmatched Schur factors give Gaussian decay in N, while the matching count and one-body terms cost $\exp(O(N\log(N+1)))$. Cauchy's inequality on the common disk gives the jets of G_N. The product kernel $(1+Z)/(1-Z)$ is estimated separately below. □

3.2 All indices in the differentiated tail

The tail estimate has two ranges. Large particle orders have enough independent angles to integrate the denominator bound directly. The remaining finite range uses the Vandermonde zero at the collision. Put $Q_0(q) = \sum_i \operatorname{dist}(q_i, 0)^2$. Equation (3.3) implies

$$|1 - Z| \ge 1 - |Z| \ge c \min\{1, N\varepsilon + Q_0(q)\}.$$

All fixed parameter derivatives of γ are bounded. Differentiating the exponential and rational factor consequently gives, for every fixed j and $0 \le r \le j$,

$$\left|\partial_u^r \frac{1+Z}{1-Z}\right| \le C_j N^j \big(1 + Q_0(q)^{-j-1}\big). \tag{3.5}$$

For $N - 1 \ge 2j + 3$, retain just the first $2j+3$ independent angle squares in Q_0. Their inverse $(j+1)$st power is integrable in $2j+3$ real dimensions, with a constant independent of N; all the unused normalized angular measures have mass one. Lemma 3.1 therefore implies

$$|\partial_u^j C_N(\mathrm{i}\varepsilon)| \le \frac{C_j N^j}{N!} \exp\{-\kappa N^2 + CN\log(N+1)\}, \qquad N \ge 2j+4. \tag{3.6}$$

This sums uniformly in ε.

For any remaining fixed N, the only real configuration at which $|Z| = 1$ is $q_1 = \cdots = q_N = 0$. Near it, set $q_N = -\sum_{i<N} q_i$, $Q = \sum_i q_i^2$ and $\Delta(q) = \prod_{i<j}(q_i - q_j)$. The numerator and every fixed parameter jet have the factor Δ^2. In radial coordinates the worst jth derivative is bounded by an integrand proportional to

$$r^{N(N-1)} r^{N-2} (\varepsilon + r^2)^{-j-1}\, \mathrm{d}r.$$

Its limiting exponent is $N^2 - 2j - 4$. If n is the first resonant index of the chosen parity, take $k = (n^2-1)/2$ for odd n or $k = n^2/2 - 1$ for even n. For every larger same-parity N and $j \le k$

this exponent is greater than -1. Combining the finitely many such indices below $2j+4$ with (3.6) proves

$$\sum_{\substack{N>n\\ N \text{ of the selected parity}}} |\partial_u^j C_N(\mathrm{i}\varepsilon)| = O_j(1), \qquad 0 \le j \le k. \tag{3.7}$$

The finite collision estimates and the large-index majorant together control the complete differentiated tail.

3.3 The first amplitude of the complete form factor

With the tail controlled, it remains to compute a nonzero contribution at the first resonant index. The independent-angle measure is kept throughout the Gaussian period calculation, so its normalization feeds directly into the singular coefficient. Choose $\phi = 2\pi a/p_0$ with p_0 an odd prime and $1 \le a \le (p_0-1)/2$. Then the first odd resonance is $n = p_0$ and the first even resonance is $n = 2p_0$. For all lower indices of the selected parity, the only possible unit-modulus product configuration has $Z = e^{-\mathrm{i}N\phi} \ne 1$, so those entire form factors are holomorphic on a neighborhood of the selected point.

In independent-angle coordinates the Gaussian Vandermonde period is

$$G_n^{\mathrm{Gauss}} := \int_{q_n = -\sum_{i<n} q_i} e^{-Q(q)} \Delta(q)^2 \, \mathrm{d}q_1 \cdots \mathrm{d}q_{n-1} = \frac{n!\,\pi^{(n-1)/2} P_n}{\sqrt{n}}, \qquad P_n = \prod_{r=1}^{n-1} \frac{r!}{2^r}. \tag{3.8}$$

The ordinary n-variable Gaussian Vandermonde integral is $n!\pi^{n/2}P_n$, by the determinant formula and the Gaussian norms of monic Hermite polynomials. Put $x_i = q_i + m$ with $\sum q_i = 0$. The coordinate Jacobian is n, while integration in m gives $\sqrt{\pi/n}$. Dividing by their product proves (3.8). Euclidean surface measure on the zero-sum hyperplane is $\sqrt{n}$ times the coordinate measure here.

Write $s_\phi = \sin\phi$ and $\alpha = (n^2-1)/2$. The singular part of the complete first form factor, on its unique collision chart, has leading expression

$$\frac{2\mathrm{i}^{1-n^2} B(0)^{n^2/2}}{n!(2\pi)^{n-1} 2^{n(n-1)} s_\phi^{n^2-1}} \int \frac{\Delta(q)^2}{nu + (B(0)/2)Q(q)} \, \mathrm{d}q_1 \cdots \mathrm{d}q_{n-1}. \tag{3.9}$$

This follows directly from

$$\gamma(q;u) = \mathrm{i}\phi - \frac{\mathrm{i}}{s_\phi}\left(u + \frac{B(0)}{2} q^2\right) + O(u^2 + uq^2 + q^4)$$

and the sine divided differences in the numerator. The complement of this unique collision chart in the fixed-n torus integral is analytic. The displayed local period therefore gives the leading coefficient of the complete form factor.

Differentiate (3.9) to the specified order and scale $q = \sqrt{\varepsilon}\, v$ on $u = \mathrm{i}\varepsilon$. The coefficient of Q in γ has positive real part, so the Gaussian/Laplace period is well-defined with its prescribed branch. The radial tails after the top derivative have radial integrands $O(R^{-3})$ for odd n and $O(R^{-2})$ for even n; their integrated tails are respectively $O(R^{-2})$ and $O(R^{-1})$. This gives dominated convergence. Terms with a higher numerator degree or an extra u or Q factor in the denominator expansion are smaller after multiplication by the indicated singularity scale; splitting at a fixed

small chart radius and then at a fixed scaled radius proves this without an unbounded formal expansion. Using (3.8) and the Laplace integral yields

$$\lim_{u\to 0,\ u=\mathrm{i}\varepsilon} u\,\partial_u^k C_n(u) = \frac{2^{1-k}\sqrt{B(0)}P_n n^{k-3/2}}{\pi^{(n-1)/2}\sin^{2k}\phi} \neq 0, \qquad k=\frac{n^2-1}{2},\quad n \text{ odd}, \tag{3.10}$$

$$\lim_{u\to 0,\ u=\mathrm{i}\varepsilon} u^{1/2}\partial_u^k C_n(u) = -\mathrm{i}\frac{2^{1-\alpha}\sqrt{B(0)}P_n n^{k-1}}{\pi^{(n-2)/2}\sin^{n^2-1}\phi} \neq 0, \qquad k=\frac{n^2}{2}-1,\quad n \text{ even}. \tag{3.11}$$

Here $u^{1/2}=e^{\mathrm{i}\pi/4}\sqrt{\varepsilon}$, and $\sqrt{B(0)}$ is fixed by the beta-period continuation from $\Re(-\mathrm{i}B(0)/(2s_\phi))>0$. For even n this square root arises from evaluating the period with the original integer power of B.

For the odd physical HT coefficients the square-root convention is particularly explicit. On Ω_H, both $\cos y$ and $\cos(\lambda y)$ are positive, and $\sin y$ and $\sin(\lambda y)$ have the same sign. Consequently

$$\Re(T\overline{S}) = \sinh x\sinh(\lambda x)\cos y\cos(\lambda y)+\cosh x\cosh(\lambda x)\sin y\sin(\lambda y)>0$$

away from the origin, where B has its positive removable value λ. Thus $\Re B>0$ throughout the HT lens; its physical square root is the principal branch. The same is true after axis exchange, since $\Re(1/B)>0$. This fixes the odd amplitudes on both transfer-axis charts without an additional sign assumption.

The lower derivatives of the even first term are bounded. For the odd first term, the $(k-1)$st derivative can be logarithmic, and all lower derivatives are still $o(u^{-1})$. Together with (3.7), these lower-derivative bounds show that the nonzero leading singularity persists in the complete sum. Multiplication by the analytic physical prefactor and by $\beta=z/(2J_x)$ preserves it: both are nonzero at a regular selected point, and the product-rule contributions from lower derivatives are smaller. A nonramified local change from u to z also preserves the nonzero top coefficient.

3.4 All low-temperature signs; the high-temperature distinction

At a negative endpoint put $q=\pi+q'$. Then $W=b+u+(-B)(1-\cos q')$. For even N, the constraint $\sum q_i=0 \pmod{2\pi}$ remains exactly the same constraint for q_i', and $B^{N^2/2}=(-B)^{N^2/2}$ because $N^2/2$ is even. Thus the whole even argument, including its tail and amplitude, applies to both endpoint signs. The other pair of edges is handled by exchanging the axes in the symmetric double contour. Rational phase values of the form above are dense on every regular arc. Section 2 then supplies the density needed on all of $\partial\Omega_L$.

For odd N, the negative endpoint has a different constraint: $\sum_i(q_i-\pi)$ is an odd multiple of π. In particular, with representatives $\delta_i=q_i-\pi\in[-\pi,\pi]$,

$$\sum_i \delta_i^2 \geq \frac{\pi^2}{N}. \tag{3.12}$$

There is no exact odd all-endpoint collision there. The same Gaussian numerator bound and (3.12) imply that the sum of every fixed parameter derivative is bounded as the negative edge is approached. This proves existence of all boundary jets, but gives no positive radius for their Taylor series. Sections 4–5 determine the growth of those jets and prove zero Taylor radius on a dense rational-phase family.

4 An exact pole normal form at a negative HT edge

At a negative spectral edge, the odd-particle momentum constraint forces the nearest collisions into two nonzero winding sectors. We locate their exact parameter poles and transform each pole into a quadratic normal form while preserving the coincidence divisors. The transformed density then yields a beta integral with estimates uniform in particle number. Keeping the full Vandermonde degree is essential for the high Taylor coefficients used in Section 5.

4.1 Coordinates, the pole equation, and the first quadratic form

The pole is centered at the nearest momentum-compatible collision. The constrained coordinates separate that center from fluctuations with zero total momentum.

Fix a regular negative endpoint and, if necessary, exchange the two lattice axes. In the notation of the angular representation, set

$$t = \frac{C + B - b}{B}, \qquad C = A/S, \qquad W(\pi + \delta; t) = b + B(t)\big(t - 1 + \cos\delta\big), \qquad b = \cos\phi. \tag{4.1}$$

Here $t = 0$ is the boundary point, the pullback is nonramified, and $B(t)$ is holomorphic and nonzero on a fixed disk. We first take the conjugation choice

$$0 < \phi < \pi, \qquad \Im B(0) < 0, \qquad \frac{\phi}{2\pi} = \frac{r}{p}, \quad (r,p) = 1, \quad p \text{ odd}. \tag{4.2}$$

The opposite sign is obtained by conjugation. Write $s = \sin\phi$ and choose the analytic branch

$$g(t,\delta) = \operatorname{arcosh} W(\pi + \delta; t), \qquad g(0,0) = \mathrm{i}\phi. \tag{4.3}$$

Because $b \in (-1,1)$, this branch is holomorphic on a fixed bidisk in (t,δ). It is even in δ, and g_t is bounded away from zero after shrinking the bidisk.

Let n tend to infinity through odd multiples of p. In the winding $+1$ chart of the momentum constraint, introduce

$$d_n = \frac{\pi}{n}, \qquad \tau_n = 1 - \cos d_n, \qquad B_n = B(\tau_n), \qquad \delta_i = d_n + v_i, \qquad V_n = \left\{ v \in \mathbb{C}^n : \sum_i v_i = 0 \right\}. \tag{4.4}$$

The integration measure on the real part of V_n is $\mathrm{d}v_1 \cdots \mathrm{d}v_{n-1}$, with $v_n = -\sum_{i<n} v_i$. Use two distinct quadratic quantities:

$$Q(v) = \sum_i v_i^2, \qquad q(v) = \sum_i |v_i|^2. \tag{4.5}$$

Use the holomorphic form Q in identities and the squared norm q in estimates. Put

$$c_n = \frac{\mathrm{i}B_n}{2s}, \qquad h_n = \cos d_n - \frac{B_n b}{s^2}\sin^2 d_n = 1 + O(n^{-2}). \tag{4.6}$$

Direct differentiation at the centre gives

$$g(\tau_n, d_n) = \mathrm{i}\phi, \qquad g_t(\tau_n, d_n) = -2c_n, \qquad \frac{1}{2} g_{\delta\delta}(\tau_n, d_n) = c_n h_n. \tag{4.7}$$

Since $n\phi$ is a multiple of 2π, the relevant pole equation is

$$\Gamma_n(t, v) := \sum_i \big(g(t, d_n + v_i) - \mathrm{i}\phi\big) = 0. \tag{4.8}$$

Lemma 4.1 (Uniform implicit pole)**.** *There are $r_0 > 0$, n_0, and constants independent of n such that, for $n \geq n_0$, equation* (4.8) *has a unique permutation-invariant holomorphic solution $t = T_n(v)$ on $v \in V_n$, $\|v\|_2 < r_0$, with*

$$T_n(0) = \tau_n, \qquad |T_n(v) - \tau_n| \leq Cq(v)/n, \tag{4.9}$$

$$\|\nabla_{V_n} T_n(v)\|_2 \leq C\sqrt{q(v)}/n, \qquad \left\|D^2 T_n(v)\right\| \leq C/n. \tag{4.10}$$

All gradients and transposes in holomorphic identities use the complex bilinear extension of the Euclidean structure on the real zero-sum plane.

Proof. Subtract the constant and linear terms of $g(t, d_n + v_i)$ in v_i. The linear terms sum to zero, the remaining sum is bounded by $Cq(v)$, and $\Gamma_{n,t}$ has modulus at least cn on a sufficiently small common neighborhood. The quantitative implicit-function theorem gives the first estimate and a single holomorphic solution on a ball whose radius is independent of n. Uniqueness gives permutation invariance. Differentiation along V_n gives

$$\nabla_{V_n} T_n = -\frac{P\big(g_\delta(T_n, d_n + v_i)\big)_{i=1}^n}{\sum_i g_t(T_n, d_n + v_i)}, \qquad P = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T.$$

The numerator has norm at most $C\|v\|_2$ and the denominator has modulus at least cn. A second differentiation gives the stated Hessian bound. The original fixed bidisk supplies every analytic derivative bound, so the resulting neighborhood is uniform in dimension. □

4.2 An exact separable energy identity

A second-order divided difference isolates the fluctuation energy. Summing it on the zero-mean plane removes the linear part and yields the exact pole identity used by the coordinate changes.

Define the following removable divided differences:

$$a_n(t) = \frac{g(t, d_n) - \mathrm{i}\phi}{t - \tau_n}, \tag{4.11}$$

$$E_n(t, v) = g(t, d_n + v) - g(t, d_n) - g_\delta(t, d_n)v, \tag{4.12}$$

$$H_n(t, v) = -\frac{2E_n(t, v)}{h_n a_n(t) v^2}. \tag{4.13}$$

The function a_n is uniformly bounded away from zero. Evenness of g in its second argument implies

$$H_n(\tau_n, 0) = 1, \tag{4.14}$$

$$|H_n(t, v) - 1| \leq C\big(|t - \tau_n| + |d_n||v| + |v|^2\big), \tag{4.15}$$

$$|\partial_v H_n(t, v)| \leq C(|d_n| + |v|), \qquad |\partial_v^2 H_n| + |\partial_t H_n| \leq C. \tag{4.16}$$

At the pole, summing (4.12) and using $\sum_i v_i = 0$ gives the exact identity

$$\boxed{T_n(v) - \tau_n = \frac{h_n}{2n} \sum_i v_i^2 H_n(T_n(v), v_i).} \tag{4.17}$$

We next turn its right-hand side into a quadratic form. Every change of variable below is holomorphic and permutation equivariant. In particular, the construction never divides by $Q(v)$, which has a nontrivial complex null cone.

4.3 The projected map and the Vandermonde divisors

The first map absorbs the scalar divided difference into each coordinate and then projects back to the zero-mean plane. Its pairwise divided differences track the Vandermonde factor through the change of variables.

Take the square root of H_n near one and set

$$U_i(v) = v_i\sqrt{H_n(T_n(v), v_i)}, \qquad w = G_n(v) := PU(v). \tag{4.18}$$

On a sufficiently small fixed ball in V_n,

$$\|DG_n - I\| \le C(|d_n|\,\|v\|_\infty + \|v\|_\infty^2 + q(v)/n) < \frac{1}{4}. \tag{4.19}$$

Thus G_n is uniformly biholomorphic on a smaller fixed ball, and its inverse has bounded derivative there. This follows directly by the uniform inverse-function estimate, or by contraction applied to $v = w - (G_n(v) - v)$. Both maps commute with coordinate permutations.

Let $\Delta(v) = \prod_{i<j}(v_i - v_j)$. Because the same $T_n(v)$ appears in every component, each pair satisfies

$$\frac{U_i - U_j}{v_i - v_j} = 1 + R_{ij}(v), \tag{4.20}$$

where the quotient extends holomorphically across $v_i = v_j$ and

$$|R_{ij}| \le C(|T_n - \tau_n| + |d_n|(|v_i| + |v_j|) + |v_i|^2 + |v_j|^2). \tag{4.21}$$

Indeed, hold $T_n(v)$ fixed and take the one-variable divided difference of $z\sqrt{H_n(T_n(v), z)}$. Projection subtracts a common mean and does not change component differences. Sum logarithms of the near-one factors before applying a norm estimate. Equations (4.9) and (4.21) give

$$\begin{aligned}\left|\log\frac{\Delta(w)}{\Delta(v)}\right| &\le C\left(n^2|T_n - \tau_n| + n|d_n|\sum_i |v_i| + nq(v)\right)\\ &\le C(\sqrt{nq(v)} + nq(v)).\end{aligned} \tag{4.22}$$

The quotient here means its holomorphic extension through the coincidence divisors.

For the Jacobian, the compressed diagonal and rank-one structure gives a trace-norm estimate sharper than a dimension times operator-norm bound. The derivative of G_n is a compressed diagonal matrix plus one rank-one matrix. The diagonal perturbation has trace norm at most

$$C\left(|d_n|\sum_i |v_i| + q(v) + n|T_n - \tau_n|\right).$$

The rank-one trace norm is at most $Cq(v)/n$, by (4.10). Since its operator norm is below $1/4$, the convergent matrix logarithm yields

$$|\log\det_{V_n} DG_n| \le C(|d_n|\sqrt{nq(v)} + q(v)). \tag{4.23}$$

The determinant on V_n is also the change-of-variable determinant in the constrained coordinates $v_1, \dots, v_{n-1}$.

4.4 The mean correction without division by a quadratic form

Projection leaves a scalar mean correction in the energy identity. The Hadamard representation expresses that correction as a bilinear form, and a rank-one square root incorporates it into the quadratic energy. This construction also works on the complex null cone.

Express v as a function of w and put

$$\mu_n(w) = \frac{1}{n}\sum_i U_i(v(w)). \tag{4.24}$$

The common linear coefficient of U_i drops out of this sum. More explicitly, at fixed t the expansion has the form

$$U_i = a_0(t)v_i + a_1(t)v_i^2 + a_2(t, v_i)v_i^3, \qquad a_1(t) = O(d_n),$$

with uniformly bounded analytic coefficients. Consequently, with $q = q(w)$,

$$|\mu_n(w)| \le \frac{C}{n}(|d_n| + \sqrt{q})q, \tag{4.25}$$

$$\|\nabla\mu_n(w)\|_2 \le \frac{C}{n}(|d_n| + \sqrt{q})\sqrt{q}, \tag{4.26}$$

$$\left\|D^2\mu_n(w)\right\| \le \frac{C}{n}(|d_n| + \sqrt{q}). \tag{4.27}$$

To obtain these bounds, differentiate the displayed expansion and use $\sum v_i = 0$, the estimates for T_n, and the bounded inverse of G_n. Terms from differentiating T_n have an additional factor $1/n$. The second derivative of G_n^{-1} is bounded by $C(|d_n| + \|w\|_2)$, so composition preserves (4.27).

The holomorphic Hadamard formula on V_n gives

$$\mu_n(w) = w^T\ell_n(w), \qquad \ell_n(w) = \int_0^1 \nabla\mu_n(\theta w)\,\mathrm{d}\theta \in V_n. \tag{4.28}$$

The field ℓ_n is permutation equivariant and satisfies

$$\|\ell_n\|_2 \le \frac{C}{n}(|d_n| + \sqrt{q})\sqrt{q}, \qquad \|D\ell_n\| \le \frac{C}{n}(|d_n| + \sqrt{q}). \tag{4.29}$$

In particular,

$$\ell_{n,i} - \ell_{n,j} = (w_i - w_j)L_{ij}(w), \qquad |L_{ij}| \le \frac{C}{n}(|d_n| + \sqrt{q}). \tag{4.30}$$

For the quantitative assertion, compare w to the vector obtained by exchanging its i, j entries. Equivariance identifies the corresponding output difference, and the derivative bound along the straight segment gives the displayed inequality. Holomorphic divisibility follows by the one-variable divided-difference formula, including on the hyperplane of coincident entries.

Now define

$$R_n(w) = \left(I + n\ell_n(w)\ell_n(w)^T\right)^{1/2} w = w + a_n^\sharp(w)\mu_n(w)\ell_n(w), \tag{4.31}$$

$$a_n^\sharp(w) = \frac{n}{1 + \sqrt{1 + n\ell_n(w)^T\ell_n(w)}}. \tag{4.32}$$

These are analytic matrix square roots near the identity, with transposes for the bilinear form Q. The matrix perturbation has operator norm at most $C(|d_n| + \sqrt{q})^2 q/n$, so R_n is uniformly biholomorphic on a smaller fixed ball. Exact multiplication gives

$$Q(R_n(w)) = Q(w) + n\mu_n(w)^2 = \sum_i U_i(v)^2. \tag{4.33}$$

The identity holds on the complex null cone because the construction uses an analytic matrix square root near the identity.

By (4.30), the relative correction to each difference $w_i - w_j$ under R_n is at most $C(|d_n| + \sqrt{q})^2 q/n$. Its summed logarithmic cost over all pairs is therefore at most Cnq on the fixed ball. Differentiating (4.31) gives a trace-norm bound $C(|d_n| + \sqrt{q})^2 q$ for $DR_n - I$, and hence the same bound for its logarithmic Jacobian. The pairwise and trace-norm estimates thus avoid an additional $n^2 \|v\|_\infty^2$ cost in the density estimate.

Lemma 4.2 (Exact symmetric normal form)**.** *On a common fixed ball in V_n, for all sufficiently large admissible n, the permutation-equivariant biholomorphism*

$$\Psi_n(v) = \sqrt{h_n}\, R_n(G_n(v)), \qquad D\Psi_n(0) = \sqrt{h_n} I, \tag{4.34}$$

with the square root near one, satisfies

$$\boxed{T_n(v) = \tau_n + \frac{Q(\Psi_n(v))}{2n}.} \tag{4.35}$$

It preserves every coincidence divisor. After extracting the constant factor $h_n^{-\alpha_n}$, where $\alpha_n = (n^2-1)/2$, the Jacobian–Vandermonde ratio has a holomorphic logarithm bounded by

$$C(\sqrt{nq} + nq). \tag{4.36}$$

The constants are independent of n.

Proof. Combine (4.17) with (4.33). The divided-difference and Jacobian bounds above give (4.36), also for the inverse map after reducing the common ball. The degree of Δ^2 is $n(n-1)$ and the dimension is $n-1$. Scaling by $\sqrt{h_n}$ therefore extracts $h_n^{-[n(n-1)+(n-1)]/2} = h_n^{-\alpha_n}$. □

4.5 The complete form-factor density at the pole

The coordinate identity alone is insufficient for a uniform coefficient estimate: the full residue density must be compared with its value at the collision center. We include the angular measure, factorial, and physical square-root convention before making that comparison.

In this chart include the constrained measure normalization in the numerator:

$$F_n(t,v) = \frac{B(t)^{n^2/2}}{n!(2\pi)^{n-1}} \prod_i \frac{1}{\sinh g(t, d_n + v_i)} \prod_{i<j} \left[\frac{\sin((v_i - v_j)/2)}{\sinh((g(t, d_n + v_i) + g(t, d_n + v_j))/2)} \right]^2. \tag{4.37}$$

The square-root sign in the odd power of B is the physical one. The global kernel is $\coth(\Gamma_n/2)$ on this resonant chart. Its pole contributes the Taylor coefficient density

$$\frac{\rho_n(v)}{T_n(v)^{j+1}}, \qquad \rho_n(v) = -\frac{2F_n(T_n(v), v)}{\Gamma_{n,t}(T_n(v), v)}. \tag{4.38}$$

The minus sign comes from expanding $(t - T_n)^{-1}$ at $t = 0$. The quotient of $\rho_n(v)$ by $\Delta(v)^2$ is holomorphic and nonzero. At the centre its value is

$$\rho_{n,0} = -\frac{2B_n^{\alpha_n - 1/2}}{n\, n!(2\pi)^{n-1} 2^{n(n-1)} s^{2\alpha_n}}. \tag{4.39}$$

Lemma 4.3 (Uniform full-density estimate)**.** *In the coordinates $w = \Psi_n(v)$, the form $\rho_n(v)\,\mathrm{d}v_1 \cdots \mathrm{d}v_{n-1}$ is*

$$\rho_{n,0} h_n^{-\alpha_n} \Delta(w)^2 (1 + \mathcal{E}_n(w))\,\mathrm{d}w_1 \cdots \mathrm{d}w_{n-1}, \tag{4.40}$$

where, uniformly in n on a common fixed ball,

$$|\mathcal{E}_n(w)| \le \exp\Big\{ C\big(\sqrt{nq(w)} + nq(w)\big) \Big\} - 1. \tag{4.41}$$

Proof. Divide the entire density by its centre value and by Δ^2 before estimating it. Every remaining analytic factor is nonzero on a common small ball, so its logarithm is available there. The power $n^2/2$ of $B(T_n)$ costs $Cn^2|T_n - \tau_n|$. For the one-body phases,

$$\sum_i |g(T_n, d_n + v_i) - \mathrm{i}\phi| \le C \left(|d_n| \sum_i |v_i| + q(v) + n|T_n - \tau_n| \right).$$

The sum over all pair-denominator logarithms costs at most Cn times this last quantity. The sine divided differences cost $C\sum_{i<j} |v_i - v_j|^2 \le Cnq(v)$. One-body denominators have the smaller single-index bound. Finally, cancellation of the linear term under $\sum v_i = 0$ gives

$$\frac{\Gamma_{n,t}(T_n(v), v)}{n g_t(\tau_n, d_n)} = 1 + O(q(v)/n).$$

Thus the logarithmic cost of the complete residue density is at most

$$C \left(n^2|T_n - \tau_n| + n|d_n| \sum_i |v_i| + nq(v) \right) \le C\big(\sqrt{nq(v)} + nq(v)\big).$$

Combining this bound with Lemma 4.2 gives the stated estimate relative to the center density, with the displayed logarithmic error. □

4.6 The real cycle, its deformation, and the boundary remainder

The deformation gives a real ball in the pole coordinates and a boundary tube. The tube and the pole-subtracted term satisfy the analytic-remainder estimate below.

Lemma 4.4 (Cycle and analytic remainder)**.** *Localize to the winding $+1$ real chart $\sum_i \delta_i^2 < \sigma$, with fixed sufficiently small $\sigma > 0$. Its domain in $V_n \cap \mathbb{R}^n$ is the actual ball $Q(v) < \sigma - \pi^2/n$. One may deform its pole integral to a real ball in the exact coordinates of Lemma 4.2. The boundary contribution and the pole-subtracted analytic remainder have Taylor coefficients bounded by*

$$(Cn)^j \exp\{-\kappa n^2 + Cn \log(n+1)\}, \qquad j \ge 0, \quad n \ge n_0, \tag{4.42}$$

where $C, \kappa > 0$ are independent of n, j. The same holds for the winding -1 chart.

Proof. First choose a fixed ball on which Ψ_n and its inverse estimates hold. Uniform near-identity bounds show that its image contains a fixed smaller ball. Next choose the real integration ball sufficiently small that its image and the straight homotopy between it and the original ball lie in this common inverse-map domain. This order of choices ensures that the transformed form is defined on the actual deformation chain. A fixed smaller ball can be used instead of the full ball $Q < \sigma - \pi^2/n$; the intervening shell obeys the same estimate below.

For small enough radius and large enough n,

$$\|\Psi_n(v) - v\|_2 \le \eta \|v\|_2, \qquad \eta < 1/4.$$

If v is real and $0 \le u \le 1$, then

$$\Re Q((1-u)v + u\Psi_n(v)) \ge (1 - 2\eta - \eta^2)Q(v) > 0. \tag{4.43}$$

Consequently the denominator $\tau_n + Q(w)/(2n)$ cannot vanish on this homotopy. Stokes' theorem for the holomorphic top-degree form changes the integral only by its boundary tube. On that tube $\Re Q(w) \ge c\sigma$, and hence

$$\left|\tau_n + \frac{Q(w)}{2n}\right| \ge \frac{c_\sigma}{n}.$$

The particle bound on this genuinely complex chain follows from the exact matched-Pfaffian identity. All angles remain in a fixed small complex neighborhood of π, and all spectral roots remain close to $\exp(-\mathrm{i}\phi)$. Every z-Schur factor therefore has a common bound $\rho_0 < 1$, while every canceled complete pair is bounded. Expanding the odd augmented Pfaffian gives $\exp\{-\kappa n^2 + O(n\log(n+1))\}$ after the one-body and factorial factors are included. No assertion that the roots have modulus at most one on the complex deformation is required. The factor $\Gamma_{n,t}^{-1}$ costs at most C/n. The maps, inverse maps, sphere volumes, and boundary parameterization cost at most $\exp(O(n\log(n+1)))$. This proves (4.42) for the boundary tube and shell.

It remains to put the pole-subtracted analytic term on the same scale. Let C_T bound $n|T_n(v)|/\sigma$ on the local ball. After shrinking σ, choose a fixed $c_{\mathrm{pole}} > 2C_T\sigma$ with $C_\Gamma(c_{\mathrm{pole}} + \sigma) < \pi$, where C_Γ is a uniform bound for the phase increment. Thus $T_n(v)$ lies inside $|t| < c_{\mathrm{pole}}/(2n)$. On $|t| < c_{\mathrm{pole}}/n$, Γ_n has only the displayed zero, $\Gamma_n/(t - T_n)$ has modulus at least $c_1 n$, and $|\Gamma_n| < \pi$. Thus $\coth(\Gamma_n/2) - 2/\Gamma_n$ is analytic there. Divide $2F_n$ by $\Gamma_n/(t - T_n)$, subtract its value at T_n, and divide the remaining analytic difference by $t - T_n$. The resulting analytic remainder has the same matched-Pfaffian bound, with polynomial factors in n absorbed into $\exp(O(n\log(n+1)))$. Cauchy's estimate on a disk of radius $c_{\mathrm{pole}}/(4n)$ gives (4.42). This local subtraction disk is chosen independently of the smaller exterior disk in Section 5. Reflection $\delta \mapsto -\delta$ treats the other winding. □

4.7 The exact beta period and the uniform two-winding formula

The constrained Gaussian period fixes the normalization of the radial beta integral. Its exact gamma ratio will be used both for a relative error bound and for the later comparison of neighboring particle indices.

Write

$$\alpha_n = \frac{n^2 - 1}{2}, \qquad P_n = \prod_{r=1}^{n-1} \frac{r!}{2^r}. \tag{4.44}$$

The constrained Gaussian period is

$$\mathcal{G}_n := \int_{V_n \cap \mathbb{R}^n} e^{-Q(v)} \Delta(v)^2 \, \mathrm{d}v_1 \cdots \mathrm{d}v_{n-1} = \frac{n! \pi^{(n-1)/2} P_n}{\sqrt{n}}. \tag{4.45}$$

Expand the Vandermonde determinant in monic Hermite polynomials, whose squared norms are $\sqrt{\pi}\, 2^{-r} r!$. The resulting unconstrained integral is $n!\pi^{n/2}P_n$. Separating the mean coordinate

has Jacobian n in the displayed constrained coordinates and contributes $\sqrt{\pi n}$, giving (4.45). Homogeneity and the beta integral now give, when $j+1>\alpha_n$,

$$\begin{aligned}\int_{V_n\cap\mathbb{R}^n} &\frac{\Delta(w)^2\,\mathrm{d}w_1\cdots\mathrm{d}w_{n-1}}{(\tau_n+Q(w)/(2n))^{j+1}}\\ &=\mathcal{G}_n(2n)^{\alpha_n}\tau_n^{\alpha_n-j-1}\frac{\Gamma(j+1-\alpha_n)}{\Gamma(j+1)}.\end{aligned}\tag{4.46}$$

The comparison below uses the exact gamma ratio. A fixed-n replacement by $j^{-\alpha_n}$ incurs a logarithmic error of order α_n^2/j in the range of comparison.

Lemma 4.5 (Uniform beta-density error)**.** *The relative effect of the density in Lemma 4.3, after the cycle deformation, is*

$$O\left(\frac{n}{\sqrt{j}}+\frac{n^2}{j}\right)\tag{4.47}$$

uniformly as $n\to\infty$ and $n^2/j\to 0$. Extension of the unperturbed radial integral from the local ball to the whole plane has an exponentially small relative error in this regime.

Proof. Under the normalized radial beta measure set $x=Q/(2n\tau_n)$. Its density is proportional to $x^{\alpha_n-1}(1+x)^{-j-1}$, and

$$nQ\asymp x,\qquad \mathbb{E}x=\frac{\alpha_n}{j-\alpha_n}.$$

The comparison constants are uniform because $2n^2\tau_n$ stays between positive constants. On $0\le x\le\delta$ with fixed $\delta>0$, the right side of (4.41) has expectation at most $C\sqrt{\alpha_n/j}+C\alpha_n/j$, using $\mathbb{E}\sqrt{x}\le\sqrt{\mathbb{E}x}$.

To control the exponential density error outside the small radial region, use the following beta-tail comparison. The truncated ball has $x\le Ln$. For $\delta\le x\le Ln$ and $j\gg n^2$,

$$C\sqrt{x}+Cx\le\tfrac{1}{4}j\log(1+x).$$

If

$$Z(\alpha,J)=\int_0^\infty x^{\alpha-1}(1+x)^{-J-1}\,\mathrm{d}x,$$

the normalized remaining integral is at most

$$(1+\delta)^{-j/4}\frac{Z(\alpha_n,j/2)}{Z(\alpha_n,j)}\le(1+\delta)^{-j/4}4^{\alpha_n},\tag{4.48}$$

provided $\alpha_n\le j/4$. In the present problem α_n is an integer, and the last ratio is exactly

$$\prod_{r=0}^{\alpha_n-1}\frac{j-r}{j/2-r}\le 4^{\alpha_n}.$$

Thus (4.48) decreases exponentially in j uniformly when $\alpha_n/j\to 0$. The same estimate without the exponential density factor bounds the omitted tail of the unperturbed beta integral. This proves (4.47). □

Proposition 4.6 (Uniform local Taylor coefficients at a negative edge)**.** *Let $a_{n,j}^{\mathrm{loc}}$ denote the Taylor coefficient of C_n from the two local windings $+1$ and -1 described above. Uniformly for odd multiples n of the fixed odd p, as $n \to \infty$ and $n^2/j \to 0$,*

$$a_{n,j}^{\mathrm{loc}} = M(n,j)\left[1 + O\left(\frac{n}{\sqrt{j}} + \frac{n^2}{j}\right)\right] + R_{n,j}, \tag{4.49}$$

where

$$\boxed{M(n,j) = -\frac{2^{2-\alpha_n} B_n^{\alpha_n - 1/2} P_n n^{\alpha_n - 3/2}}{\pi^{(n-1)/2} s^{2\alpha_n}} h_n^{-\alpha_n} \tau_n^{\alpha_n - j - 1} \frac{\Gamma(j+1-\alpha_n)}{\Gamma(j+1)},} \tag{4.50}$$

and

$$|R_{n,j}| \le (Cn)^j \exp\{-\kappa n^2 + Cn \log(n+1)\}. \tag{4.51}$$

All constants are independent of n, j in this regime.

Proof. Multiply the centre density (4.39), the factor $h_n^{-\alpha_n}$, and the exact period (4.46). Reflection $\delta \mapsto -\delta$ maps the two winding charts to one another; their coefficients are equal and add, giving the factor two in (4.50). Lemma 4.5 gives the relative error, and Lemma 4.4 gives (4.51). □

Remark 4.7. The phase of $B_n^{\alpha_n - 1/2}$ varies quadratically with n. The next section compares the magnitudes of the competing terms. The uniform estimate (4.49) applies throughout $\log j \le n \le \sqrt{j}/(\log j)^{1/4}$.

5 Negative high-temperature edges: boundary jets and the complete odd sum

We now use the local coefficient formula to prove nonanalyticity of the complete odd sum. First we identify its boundary jets by absolute summation. We then separate analytic remainders, higher windings, and the small and large particle ranges. A discrete concavity estimate selects Taylor orders at which one intermediate particle index dominates. The resulting growth contradicts any positive Taylor radius.

5.1 Setting and the existence of the boundary jets

Fix a regular negative endpoint. After complex conjugation if necessary, the local analytic data can be written

$$W(t, \pi + \delta) = b + B(t)\big[t - (1 - \cos\delta)\big], \qquad b = \cos\phi, \quad \phi = \frac{2\pi r}{p} \in (0, \pi), \tag{5.1}$$

where $p \ge 3$ is odd, $(r,p) = 1$, B is holomorphic and nonzero, $\Im B(0) < 0$, and t is a nonramified local temperature coordinate. The physical inward direction is $t = -\varepsilon$, $\varepsilon > 0$. The branch $\cosh\gamma(q;t) = W(t,q)$ has $\gamma(\pi;0) = \mathrm{i}\phi$. All assertions below concern this branch. At a regular negative edge, the earlier root construction gives a common branchfree parameter neighborhood for the entire real angular circle.

To keep its normalization visible, write the complete coefficient as

$$C_n(t) = \int_{\mathcal{T}_n} H_n(t,q) \coth\!\left(\frac{\Gamma_n(t,q)}{2}\right) \mathrm{d}q, \qquad \Gamma_n = \sum_{i=1}^{n} \gamma(q_i;t), \tag{5.2}$$

where $\mathcal{T}_n$ is the zero-total-momentum torus, $q_n = -\sum_{i<n} q_i$ modulo 2π, the measure $\mathrm{d}q$ is the ordinary measure in the $n-1$ independent angles, and

$$H_n(t,q) = \frac{B(t)^{n^2/2}}{n!(2\pi)^{n-1}} \prod_{i=1}^{n} \frac{1}{\sinh \gamma(q_i;t)} \prod_{i<k} \left[\frac{\sin((q_i-q_k)/2)}{\sinh((\gamma(q_i;t)+\gamma(q_k;t))/2)}\right]^2. \tag{5.3}$$

The branch of $B^{1/2}$ is the one in the physical representation. The normalization in (5.3) is retained throughout this section. The physical susceptibility differs from the complete odd sum by the analytic prefactor already identified in the physical representation; that prefactor is nonzero at a regular endpoint.

Set

$$S(q) = \sum_{i=1}^{n} \operatorname{dist}(q_i,\pi)^2.$$

The spectral geometry and the matched Schur–Pfaffian estimate imply, with constants independent of n,

$$\Re\gamma(q;0) \geq c_0 \operatorname{dist}(q,\pi)^2, \quad |\gamma(q;0) - \mathrm{i}\phi| \leq C \operatorname{dist}(q,\pi)^2, \tag{5.4}$$

$$\sup_{|t|<r_0,\ q\in\mathcal{T}_n} |H_n(t,q)| \leq E_n, \qquad E_n = \exp\{-\kappa n^2 + Cn\log(n+1) + C\}. \tag{5.5}$$

Choose r_0 small enough that the strict Schur bound persists on the fixed complex disk, by continuity from the edge.

For odd n, choose the shortest lifts $q_i = \pi + \delta_i$ modulo 2π. Momentum conservation gives $\sum_i \delta_i = w\pi$ with w odd. Therefore

$$S(q) \geq \frac{\pi^2}{n}. \tag{5.6}$$

This momentum gap gives an n^{-2} analytic scale for each coefficient; the scale shrinks with particle number.

Lemma 5.1 (Coefficientwise disks and boundary derivatives)**.** *There is $c>0$ such that each odd coefficient is holomorphic on $|t|<c/n^2$, agrees there with* (5.2)*, and obeys*

$$\sup_{|t|<c/n^2} |C_n(t)| \leq CnE_n, \qquad |[t^j]C_n(t)| \leq (Cn^2)^j CnE_n. \tag{5.7}$$

The complete odd sum on $t<0$ has derivatives of every fixed order at $t=0$ from that side, and its normalized boundary jets are

$$a_j = \lim_{\varepsilon\downarrow 0} \frac{1}{j!}\partial_t^j \sum_{n \text{ odd}} C_n(-\varepsilon) = \sum_{n \text{ odd}} [t^j]C_n(t). \tag{5.8}$$

The sum on the right converges absolutely for each fixed j.

Proof. The parameter derivative of each one-particle phase is bounded on a fixed disk. Thus, using (5.4) and (5.6), a sufficiently small perturbation $|t| < c/n^2$ preserves $\Re\Gamma_n \geq c'/n$. The modulus of the hyperbolic cotangent is then at most Cn. This proves the first bound; Cauchy's estimate proves the second.

For the inward real approach, (5.1) gives

$$\Re\gamma(q;-\varepsilon) \geq c\big(\varepsilon + \operatorname{dist}(q,\pi)^2\big).$$

Indeed $\Im W = (-\Im B(-\varepsilon))(\varepsilon + 1 - \cos\delta)$ is positive, and writing $\gamma = v + \mathrm{i}\psi$ expresses it as $\sinh v \sin\psi$, with ψ in a compact subinterval of $(0, \pi)$. A disk centered at $-\varepsilon$ of radius $c(\varepsilon + n^{-2})$ consequently preserves the same denominator gap, up to a fixed factor. Cauchy's estimate there bounds the jth derivative by $j!(Cn^2)^j CnE_n$, uniformly for small ε. This is summable for every fixed j. Differentiation, the boundary limit, and the particle sum therefore commute by dominated convergence. Thus every fixed boundary jet exists by absolute summation. The radius of the resulting Taylor series is the separate question addressed below. □

5.2 Analytic remainder estimates

Choose $\sigma > 0$ sufficiently small, and then a fixed $c_{\mathrm{out}} > 0$ such that $C_\Gamma c_{\mathrm{out}} < c_0\sigma/2$ and $C_\Gamma(\sigma + c_{\mathrm{out}}) < \pi/p$. Here C_Γ bounds the phase increments below. On $|t| < c_{\mathrm{out}}/n$,

$$\Re\Gamma_n \geq c_0 S - C_\Gamma c_{\mathrm{out}}, \qquad |\Gamma_n - n\mathrm{i}\phi| \leq C_\Gamma(S + c_{\mathrm{out}}). \tag{5.9}$$

If $S \geq \sigma$, the first estimate supplies a fixed positive real part by this choice of c_{out}. If $S < \sigma$ and $p \nmid n$, the second estimate supplies a fixed phase gap: the distance from $n\phi$ to $2\pi\mathbb{Z}$ is at least $2\pi/p$. Thus the whole coefficient for $p \nmid n$, and the part with $S \geq \sigma$ for any n, are holomorphic on this c_{out}/n disk with norm at most CE_n. The angular cutoff is fixed and independent of t.

For $p \mid n$, put $D_n = \Gamma_n - n\mathrm{i}\phi$. On the inner region $S < \sigma$, $|D_n| < \pi$ on the same disk if the constants are sufficiently small. The identity

$$\coth(D_n/2) = \frac{2}{D_n} + R(D_n), \qquad R \text{ holomorphic on } |D_n| < 2\pi, \tag{5.10}$$

places the R part in the same analytic remainder class.

For the remaining resonant region, subtract the exact simple parameter pole before applying the quadratic normal form. In the inner region use the unique small lifts $q_i = \pi + \delta_i$ and put $v_i = 1 - \cos\delta_i$. Let

$$d_n(t, q) = \frac{1}{n} D_n(t, q), \qquad a_0 = \frac{B(0)}{\sinh(\mathrm{i}\phi)} \neq 0.$$

On a sufficiently small fixed parameter disk,

$$\partial_t d_n(t, q) = a_0 + O(|t| + \sigma),$$

uniformly in n and in $S < \sigma$. Reducing the disk and σ makes this derivative lie within $|a_0|/4$ of a_0. The quantitative inverse-function estimate gives a unique zero $t = T_n(q) = O(S/n)$ and the factorization

$$D_n(t, q) = n(t - T_n(q))A_n(t, q), \qquad A_n(t, q) = \int_0^1 \partial_t d_n(T_n(q) + u(t - T_n(q)), q)\,\mathrm{d}u. \tag{5.11}$$

The factor A_n is analytic and uniformly bounded away from zero on a smaller fixed disk containing every such $T_n(q)$. The proof can also be obtained directly by Rouché's theorem on a circle of radius CS/n: $|d_n(0, q)| \leq CS/n$, while the derivative differs uniformly from the nonzero constant a_0.

Set $F_n = H_n/A_n$. The exact identity

$$\frac{2H_n(t, q)}{D_n(t, q)} = \frac{2F_n(T_n(q), q)}{n(t - T_n(q))} + \frac{2}{n}\frac{F_n(t, q) - F_n(T_n(q), q)}{t - T_n(q)} \tag{5.12}$$

has a holomorphic second term, including at $t = T_n(q)$. A fixed-disk Cauchy bound on $\partial_t F_n$ bounds this divided difference by CE_n. The removable divided difference supplies this bound even when the parameter approaches the pole.

Lemma 5.2 (Exact reduction to the inner pole integrals)**.** *For every $j \geq 0$ the boundary coefficient in* (5.8) *is the sum of the exact inner pole coefficients for odd n divisible by p, together with a remainder r_j satisfying*

$$|r_j| \leq \exp\left\{\frac{1}{2} j \log(j+1) + Cj + C\right\}. \tag{5.13}$$

For a resonant n, the inner pole coefficient is exactly

$$a_{j,n}^{\text{pole}} = -\int_{S<\sigma} \frac{2H_n(T_n(q), q)}{\partial_t \Gamma_n(T_n(q), q)} T_n(q)^{-j-1}\, \mathrm{d}q. \tag{5.14}$$

Proof. All removed terms are bounded on a disk c_{rem}/n, where c_{rem} is a fixed sufficiently small fraction of $\min(c_{\text{pole}}, c_{\text{out}})$. Construct the pole subtraction on its larger disk, then restrict the resulting identity to the c/n^2 disk of the original integral. Taking Taylor coefficients there gives (5.14); the minus sign is the coefficient of $(t-T)^{-1}$ at $t=0$. Every removed term is analytic on $|t| < c_{\text{rem}}/n$ with norm bounded by CE_n. Their total coefficient bound is

$$\sum_{n \geq 1} (Cn)^j E_n.$$

Absorb $Cn\log(n+1)$ into $(\kappa/2)n^2 + C'$, and optimize $j\log n - (\kappa/2)n^2$. This gives (5.13), including summability of the tail. □

5.3 Uniform bounds for the winding sectors

Two bounds are needed. The beta integral distinguishes the first windings from the higher ones when its convergence condition holds. A separate ball-mass bound controls arbitrarily large particle orders.

The exact pole is quantitatively close to the average $\bar{v} = n^{-1}\sum_i v_i$. At $t = \bar{v}$, the linear terms in the analytic expansion of $\operatorname{arcosh}(b + B(\bar{v})(\bar{v} - v_i))$ sum to zero. The same uniform inverse-function estimate as above consequently gives

$$T_n(q) = \bar{v} + O\big(\operatorname{Var}(v)\big), \qquad |T_n(q)| \geq (1 - C\sigma)\frac{S(q)}{2n}. \tag{5.15}$$

Here $\operatorname{Var}(v) = n^{-1}\sum_i (v_i - \bar{v})^2 \leq C\sigma\bar{v}$, and $\bar{v} \geq (1 - C\sigma)S/(2n)$, giving the stated lower bound for the complex pole modulus.

Write $s = \sin\phi > 0$, $\alpha_n = (n^2-1)/2$, and

$$H_{0,n} = \frac{|B(0)|^{n^2/2}}{n!(2\pi)^{n-1} 2^{n(n-1)} s^{n^2}}. \tag{5.16}$$

The complete numerator, evaluated at the pole, satisfies

$$|H_n(T_n(q), q)| \leq H_{0,n}\Delta(\delta)^2 e^{CnS}, \qquad \Delta(\delta) = \prod_{i<k}(\delta_i - \delta_k). \tag{5.17}$$

To check the constant and the dependence on n, divide every sine numerator by $(\delta_i - \delta_k)/2$ and every hyperbolic denominator by $\sinh(\mathrm{i}\phi)$. Analytic logarithms of the resulting near-one ratios are bounded by $C(\delta_i - \delta_k)^2$ and $C(|T_n| + \delta_i^2 + \delta_k^2)$, respectively. Their sums are $O(nS + n^2|T_n|) = O(nS)$. The one-body factors cost $O(S)$. Finally the logarithm of $B(T_n)^{n^2/2}/B(0)^{n^2/2}$ costs $O(n^2|T_n|)$.

This proves (5.17) relative to the exact scale (5.16); an unspecified factor raised to n^2 would be too costly for the later comparison.

In winding sector w, where w is odd, put

$$\delta_i = \frac{w\pi}{n} + \eta_i, \quad \sum_i \eta_i = 0, \quad Q = \sum_i \eta_i^2, \quad m_{n,w} = \frac{\pi^2 w^2}{n}.$$

Then $S = Q + m_{n,w}$ and $\Delta(\delta) = \Delta(\eta)$. There are at most $C\sqrt{n}$ sectors with nonempty $S < \sigma$. The constrained Gaussian integral, with the same independent-angle measure as above, is

$$I_n = \int_{\mathbb{R}^{n-1}} e^{-Q} \Delta(\eta)^2 \, \mathrm{d}\eta = \frac{\pi^{(n-1)/2} 2^{-n(n-1)/2}}{\sqrt{n}} \prod_{r=1}^{n} r!. \tag{5.18}$$

For example, separating the center of mass in the ordinary Gaussian Vandermonde integral gives a Jacobian n and a center Gaussian $\sqrt{\pi/n}$, whose product explains the constrained $1/\sqrt{n}$ factor. Homogeneity therefore gives, whenever $j + 1 > \alpha_n$,

$$\int_{\mathbb{R}^{n-1}} \frac{\Delta(\eta)^2}{(Q+m)^{j+1}} \, \mathrm{d}\eta = I_n m^{\alpha_n - j - 1} \frac{\Gamma(j + 1 - \alpha_n)}{\Gamma(j+1)}. \tag{5.19}$$

Lemma 5.3 (Higher windings and large particle numbers)**.** *For $j + 1 > \alpha_n$, the exact pole coefficient in winding w satisfies*

$$|a^{\mathrm{pole}}_{j,n,w}| \le \frac{C H_{0,n}}{n} e^{Cn\sigma} \left(\frac{2n}{1 - C\sigma} \right)^{j+1} I_n m_{n,w}^{\alpha_n - j - 1} \frac{\Gamma(j + 1 - \alpha_n)}{\Gamma(j+1)}. \tag{5.20}$$

For every resonant odd n and every $j \ge 0$, without a restriction involving α_n,

$$\log \sum_w |a^{\mathrm{pole}}_{j,n,w}| \le 2j \log n - \frac{1}{2} n^2 \log n + Cn^2 + Cj + C. \tag{5.21}$$

Empty sectors are understood to contribute zero.

Proof. The residue denominator has modulus at least cn. Combine this with (5.15) and (5.17), replace S in its exponential by σ, and extend the positive radial majorant to the whole real zero-sum plane. Equation (5.19) proves (5.20). Its exact dependence on w relative to the same expression with $w = 1$ is $|w|^{-2(j+1-\alpha_n)}$.

For the second estimate, use the simpler bound $|T_n| \ge c/n^2$ and the exact ball mass

$$\int_{Q<\sigma} \Delta(\eta)^2 \, \mathrm{d}\eta = I_n \frac{\sigma^{\alpha_n}}{\Gamma(\alpha_n + 1)}.$$

The logarithm of this ball mass is $-\frac{1}{2} n^2 \log n + O(n^2)$; this follows by applying Stirling's inequalities to (5.18) and to $\Gamma(\alpha_n + 1)$. Multiplication by $H_{0,n}$ costs $O(n^2)$. The number of windings and the remaining polynomial factors are absorbed into the displayed error. This proves (5.21) for arbitrarily large n. □

5.4 The exact first-winding model

We now collect the center density, exact beta period, and the two equal first-winding contributions into a single model. Its complex phase is retained because different particle indices need not have aligned phases.

Let

$$\tau_n = 1 - \cos(\pi/n), \quad B_n = B(\tau_n), \quad h_n = \cos(\pi/n) - \frac{B_n b}{s^2}\sin^2(\pi/n), \quad P_n = \prod_{r=1}^{n-1} r!\, 2^{-r}. \tag{5.22}$$

In particular $h_n = 1 + O(n^{-2})$. The exact symmetric pole normal form and cycle estimate of Proposition 4.6 in Section 4 give, uniformly as $n \to \infty$ through odd multiples of p and $n^2/j \to 0$,

$$a^{\mathrm{pole}}_{j,n,1} + a^{\mathrm{pole}}_{j,n,-1} = M(n,j)(1 + e_{n,j}) + b_{n,j}, \tag{5.23}$$

where

$$M(n,j) = -\frac{2^{2-\alpha_n} B_n^{\alpha_n - 1/2} P_n n^{\alpha_n - 3/2}}{\pi^{(n-1)/2} s^{2\alpha_n}} h_n^{-\alpha_n} \tau_n^{\alpha_n - j - 1} \frac{\Gamma(j + 1 - \alpha_n)}{\Gamma(j+1)}, \tag{5.24}$$

and

$$|e_{n,j}| \le C\left(\frac{n}{\sqrt{j}} + \frac{n^2}{j}\right), \qquad |b_{n,j}| \le (Cn)^j \exp\{-\kappa' n^2 + C' n \log(n+1) + C'\}. \tag{5.25}$$

The same estimate applies to the actual inner ball of radius squared $\sigma - \pi^2/n$, since these radii stay between fixed positive constants for large n. Alternatively its outer shell belongs to a c/n analytic remainder by the preceding argument.

The constants in (5.24) follow directly from the central numerator, pole derivative, and constrained beta period. At the collision $\gamma_i = \mathrm{i}\phi$, the numerator divided by Δ^2 is

$$-\frac{\mathrm{i} B_n^{n^2/2}}{n!(2\pi)^{n-1} 2^{n(n-1)} s^{n^2}}, \qquad \Gamma_t = -\frac{\mathrm{i} n B_n}{s}.$$

Multiplication by $-2/\Gamma_t$, followed by the beta integral in the exact normal form, gives half of (5.24). Reflection $\delta \mapsto -\delta$ preserves the full integrand and exchanges the first two winding sectors, giving the displayed factor of two. The factor $B_n^{\alpha_n - 1/2}$ retains its quadratic particle-dependent phase. No positivity across particle indices is being assumed.

5.5 A discrete concavity estimate with controlled errors

Dominance requires a comparison between adjacent allowed particle indices. We compute the second difference from the exact factorial, gamma, and pole-location factors, keeping the error uniform over the whole intermediate window.

Write $L(n,j) = \log|M(n,j)|$ and set $d = 2p$, the step between successive allowed odd multiples of p. Define

$$c_* = -\frac{3}{4} - \frac{3}{2}\log 2 + \log\pi + \frac{1}{2}\log|B(0)| - \log s. \tag{5.26}$$

Elementary Stirling estimates in the exact formula give

$$L(n,j) = 2j\log n - \frac{n^2}{2}\log j + c_* n^2 - j\log(\pi^2/2) + O(n\log n + n^4/j + j/n^2), \tag{5.27}$$

when $n \to \infty$ and $n^2/j \to 0$. The expansion locates the relevant scale. For the concavity estimate, we return to the exact factors so that the error remains controlled under discrete differences.

Lemma 5.4 (Exact-factor second differences)**.** *Uniformly when $n \to \infty$ and $n^2/j \to 0$,*

$$L(n+d,j) - 2L(n,j) + L(n-d,j) = d^2\left(-\frac{2j}{n^2} - \log j + 2c_*\right) + O_d(n^2/j + j/n^4 + n^{-2}). \quad (5.28)$$

In particular this difference is at most $-cd^2 \log j$ throughout the interval

$$\log j < n < \frac{\sqrt{j}}{(\log j)^{1/4}}, \quad (5.29)$$

for all sufficiently large j, whenever the adjacent indices are also in that interval.

Proof. The exact quotient of the factorial products gives

$$\begin{aligned}\Delta_d^2 \log P_n &= \sum_{k=0}^{d-1}\sum_{\ell=1}^{d} \log(n-d+k+\ell) - d^2\log 2 \\ &= d^2(\log n - \log 2) + O_d(n^{-2}).\end{aligned}$$

The offsets in the double sum add to zero, so their first-order $1/n$ terms cancel.

For integer $j > \alpha_{n+d}$, the gamma quotient can be written

$$\log\frac{\Gamma(j+1-\alpha_n)}{\Gamma(j+1)} = -\alpha_n \log j + \sum_{m=0}^{\alpha_n - 1} -\log(1-m/j).$$

The two adjacent increments of α_n have lengths $dn + d^2/2$ and $dn - d^2/2$. Their d^2 excess summands cost $O_d(n^2/j)$. Pairing the remaining summands costs the same amount, because the function $-\log(1-x/j)$ has derivative at most $1/(j-\alpha_{n+d})$ on the required interval. Thus this factor has second difference $-d^2\log j + O_d(n^2/j)$.

Finally

$$\log\tau_n = \log(\pi^2/2) - 2\log n - \frac{\pi^2}{12n^2} + O(n^{-4}),$$

while $B(\tau_n) = B(0)(1+O(n^{-2}))$ and $h_n = 1 + O(n^{-2})$ have convergent expansions in $1/n$ near zero. Computing the second differences of these explicit terms in (5.24) gives (5.28), including the error j/n^4 from the exact value of τ_n.

In (5.29), the error j/n^4 is smaller than j/n^2 by a factor at most $(\log j)^{-2}$, and n^2/j tends uniformly to zero. This gives the negative second difference throughout the intermediate range. □

5.6 Selecting an integer order with one dominating index

Choose the Taylor order by balancing the two neighbors of a prescribed allowed index. Strict concavity then places that index above both neighbors and controls the entire intermediate range.

Fix a large allowed index n and let $F_n(j) = L(n+d,j) - L(n-d,j)$, initially with real $j \asymp n^2 \log n$. The two gamma arguments differ by the integer $2dn$, so differentiation of their exact quotient gives

$$\begin{aligned}F_n'(j) &= \log\frac{\tau_{n-d}}{\tau_{n+d}} - \sum_{m=1}^{2dn}\frac{1}{j+1-\alpha_{n-d}-m} \\ &= \frac{4d}{n} - \frac{2dn}{j} + O_d(n^{-3} + n^3/j^2) > 0. \quad (5.30)\end{aligned}$$

The exact first difference, with the ordinary Stirling bounds for the finitely many factorials in each quotient, also gives

$$\frac{F_n(j)}{2d} = \frac{2j}{n} - n\log j + 2c_*n + O_d(\log n + n^3/j + j/n^3). \tag{5.31}$$

At

$$j = n^2\bigl(\log n + \tfrac{1}{2}\log\log n - c_* \pm \varepsilon\bigr)$$

this expression has opposite signs for every fixed $\varepsilon > 0$ and all sufficiently large n. Consequently there is a unique crossing in this range. Round it to the nearest integer and denote the result by j_n. Then

$$j_n = n^2\bigl(\log n + \tfrac{1}{2}\log\log n - c_* + o(1)\bigr), \qquad F_n(j_n) = O_d(1/n). \tag{5.32}$$

The selected n lies strictly inside (5.29) for these j_n.

Let $\ell_k = L(n+kd, j_n)$ for the allowed indices in that window. Lemma 5.4 says that successive differences of ℓ_k decrease by at least $cd^2\log j_n$. Equation (5.32) makes the two adjacent values equal up to $O_d(1/n)$. It follows first that $\ell_0 - \ell_{\pm1} \geq c'd^2\log j_n$, and then, by summing the decreasing first differences, that

$$\ell_k - \ell_0 \leq -c''d^2k^2\log j_n + O_d(|k|/n) \qquad (k \neq 0).$$

In particular

$$\sum_{\substack{m\neq n,\ p\mid m,\ m \text{ odd} \\ m \text{ in (5.29)}}} |M(m, j_n)| \leq |M(n, j_n)|\, O(j_n^{-c'''d^2}). \tag{5.33}$$

The modulus gap makes the comparison independent of the varying complex phases. An $o(1)$ relative error in the local model is sufficient: the error of the largest index is relative to its own modulus, while the errors at other indices multiply already smaller moduli.

5.7 Completing the all-index comparison

We can now compare every contribution with the selected model term. The low range uses coefficientwise Cauchy estimates; the high range uses the ball-mass bound. In the intermediate range, concavity controls the other indices, the winding estimate controls the remaining sectors, and the normal-form estimate controls the relative error.

For $j \to \infty$, split the odd particle indices at

$$n_{\mathrm{lo}} = \log j, \qquad n_{\mathrm{hi}} = \frac{\sqrt{j}}{(\log j)^{1/4}}.$$

The low range, including every nonasymptotic index, is bounded by Lemma 5.1:

$$\sum_{n\leq n_{\mathrm{lo}}} |[t^j]C_n| \leq \exp\{2j\log\log j + O(j)\}. \tag{5.34}$$

All c/n analytic remainders and the normal-form boundary errors have the smaller scale $\exp\{\frac{1}{2}j\log j + O(j)\}$ by (5.13) and (5.25).

For $n \geq n_{\mathrm{hi}}$, use the ball-mass bound (5.21). In the variable $x = n^2$ its controlling exponent is

$$j\log x - \tfrac{1}{4}x\log x + Cx + Cj,$$

which is decreasing for every $x \geq j/\sqrt{\log j}$ when j is sufficiently large. At the threshold it equals at most

$$j \log j - \tfrac{1}{2} j \log\log j - \tfrac{1}{4} j\sqrt{\log j} + o(j\sqrt{\log j}). \tag{5.35}$$

Its subsequent decrease makes the full infinite tail summable, with the same leading bound. This estimate handles all windings in the high range and requires no beta integral outside its convergence range.

In the intermediate range, the error in (5.23) is uniformly $O((\log j)^{-1/4}) = o(1)$. It remains to compare the higher odd windings to that model. Let $M_0(n, j)$ be the modulus of (5.24) with B_n replaced by $B(0)$, h_n by 1, and τ_n by $\pi^2/(2n^2)$. The exact formulas give

$$\left|\log \frac{M_0(n, j)}{|M(n, j)|}\right| \leq C(j/n^2 + 1) = o(j) \tag{5.36}$$

uniformly in this entire range. Equation (5.20) is bounded by $CM_0(n, j)$ times

$$e^{Cn\sigma}(1 - C\sigma)^{-j-1}|w|^{-2(j+1-\alpha_n)}.$$

Here $n = o(j)$ and $\alpha_n/j = o(1)$ uniformly. Choose the fixed σ so small that $-\log(1-C\sigma) < \frac{1}{2}\log 3$. Summing over all odd $|w| \geq 3$ then gives

$$\sum_{|w|\geq 3} |a^{\text{pole}}_{j,n,w}| \leq |M(n, j)|e^{-cj} \tag{5.37}$$

for some $c > 0$ and all sufficiently large j, uniformly over the intermediate range. Together, the low, intermediate, and high ranges now include every particle index and every winding sector.

At the selected sequence (5.32), the exact model and (5.27) yield

$$\begin{aligned} &\log |M(n, j_n)| = j_n \log j_n - j_n \log\log j_n + j_n\big(\log(4/\pi^2) - 1\big) + o(j_n), \\ &|M(n, j_n)|^{1/j_n} \sim \frac{4j_n}{e\pi^2 \log j_n}. \end{aligned} \tag{5.38}$$

For this calculation, the exact gamma ratio differs from $-\alpha_n \log j_n$ in its logarithm by $O(\alpha_n^2/j_n)$, which is $o(j_n)$ at the selected scale. The finite differences and relative estimates above use the exact ratio.

The bounds (5.34) and (5.35), and all analytic remainders, are $o(|M(n, j_n)|)$ by (5.38). The intermediate higher windings have the same property by (5.37) and (5.33). Finally the uniform relative error in (5.23), summed over that range, is at most $o(1)\sum_m |M(m, j_n)| = o(|M(n, j_n)|)$. Therefore the complete boundary coefficient satisfies

$$a_{j_n} = M(n, j_n)(1 + o(1)). \tag{5.39}$$

The complete odd-particle coefficient is therefore asymptotic to the single selected model term, despite its phase dependence on n.

Theorem 5.5 (A regular rational negative edge is singular for the full odd sum)**.** *Under the physical-branch and nonramified-coordinate hypotheses of* (5.1)*, the complete odd sum has all inward boundary derivatives at $t = 0$, but the Taylor series of those boundary derivatives has radius of convergence zero. More precisely, along the integer sequence* (5.32)*,*

$$|a_{j_n}|^{1/j_n} \sim \frac{4j_n}{e\pi^2 \log j_n} \longrightarrow \infty.$$

Consequently the complete high-temperature physical germ has no holomorphic continuation across this regular negative edge point. Multiplication by its nonzero analytic physical normalization preserves this conclusion.

Proof. The jets exist and equal the absolutely convergent coefficient sums by Lemma 5.1. Equations (5.38) and (5.39) give their stated growth and zero Taylor radius. If there were a holomorphic continuation across the point, its Taylor coefficients would be these same boundary jets and would satisfy a positive-radius Cauchy bound, a contradiction. The physical normalization can be divided out in a neighborhood because it is analytic and nonzero there. The use of the nonramified coordinate t makes this a statement about the fixed-ratio one-variable physical germ itself. □

6 Natural boundaries at unequal and equal coupling

6.1 Completing the unequal-coupling theorem

The holomorphic sums in Section 1 agree with the physical susceptibilities on the respective real intervals. The identity theorem fixes the continuations on the connected domains Ω_H and Ω_L.

For each fixed $\lambda \neq 1$, the exceptional set in Section 2 is locally discrete. Every other boundary point has a real analytic edge arc with exactly one endpoint cosine, a strictly interior second cosine, a nonzero real-linear determinant, and a nonramified holomorphic edge coordinate. Restricting that coordinate to the arc gives an open real interval. Cosines of primitive odd-prime rational angles are dense in every such interval.

On every regular LT arc, Section 3 proves a genuine singularity of the entire even-particle sum at each point of this dense family, including both endpoint signs and both transfer-axis choices. The first singular derivative has a nonzero coefficient and every higher even index is included in the absolute bounded tail. On every regular positive HT arc, the odd counterpart proves the analogous whole-sum singularity. On every regular negative HT arc, Sections 4–5 show that all fixed boundary jets exist, but a sequence of their Taylor coefficients satisfies

$$|a_{j_n}|^{1/j_n} \sim \frac{4j_n}{e\pi^2 \log j_n} \longrightarrow \infty. \tag{6.1}$$

These coefficients are those of the complete odd sum. The Gaussian bound permits taking the boundary limit through the sum for each fixed derivative. If a holomorphic extension existed, its derivatives would have these limits and its Taylor series would have positive radius, contradicting (6.1). Division by the local analytic nonzero physical prefactor proves the same assertion for $\mathcal{X}_H$, and multiplication by the nonzero β gives χ_H.

Finally suppose a holomorphic extension existed through any boundary point, including an exceptional point. A small disk of extension contains regular arcs of the same physical boundary, and hence contains a nearby point of the appropriate dense true-singularity family. Restricting the extension to that point contradicts its already proved nonremovability. The complete graph $\partial\Omega_L$ and the complete closed curve $\partial\Omega_H$ therefore are natural boundaries. This proves the unequal-coupling part of Theorem 1.1.

The conclusion concerns the boundaries of the physical complementary components of $\mathcal{S}_\lambda$. For unequal couplings $\mathcal{S}_\lambda$ has interior, while the two barriers are the explicit one-dimensional boundaries. Their obstruction follows from the complete-sum estimates and the closure argument above.

6.2 Exceptional points and dependence on the ratio

For $\lambda \neq 1$, the locally discrete collinear points, corners, elementary zeros or poles, and critical points of the edge coordinate are included by density. Section 2 treats the simultaneous imaginary-axis cosine zeros at odd-over-odd rational ratios and shows a nonzero opening at the real critical point for every unequal positive ratio. The negative-edge estimates use the holomorphic germ $B(t)$ and the exact centers $1 - \cos(\pi/n)$, with constants depending on the chosen regular point at the fixed ratio. For $\lambda = 1$ the real-linear determinant degenerates along the spectral curve, and the following proof uses the isotropic formulas.

6.3 The isotropic slice and its prior boundary theorems

The isotropic transfer is an identity of the full coefficient formulas. Set $\lambda = 1$ and $s = \sinh z$. Then $S = T = s$, $p = s^2$, $B = 1$, and

$$D(x, y; z) = s\left[s + s^{-1} - \tfrac{1}{2}(x + x^{-1}) - \tfrac{1}{2}(y + y^{-1})\right].$$

On the positive LT interval $\sqrt{p} = s$, so every factor $\sqrt{p}/D$ in (1.10) is exactly the normalized isotropic factor in [7]. The even coefficients and the full-site identity consequently agree term by term, with $\xi_L = (1 - s^{-4})^{1/4}$. In the HT interval the full angular coefficient in (1.13) has $W = s + s^{-1} - \cos q$, and its prefactor is $(1 - s^4)^{1/4}/s$. This is the normalized full-lattice odd series in [8]. In the notation of that HT manuscript its auxiliary exterior variable is $\widehat{t} = 1/s$; hence our C_N at s is exactly its $C_N(\widehat{t})$, and the reported real condition $\widehat{t} > 1$ is exactly the present physical interval $0 < s < 1$.

We invoke the following normalized isotropic results:

1. the exterior connected full-bulk result in [7, Corollary 9.3], derived there from the whole-form-factor Theorem 4.4 and the differentiated-tail Theorem 9.1;

2. the normalized HT analytic germ and unit-circle obstruction of [8], Theorem 1.1. Its physical identification requires the real full-bulk form-factor equality, absolute onsite summability and onsite normalization, addressed in Section 1 at $B = 1$.

The cited LT theorem gives, at its dense prime family, $N_0 = 2p$ and $k = N_0^2/2 - 1$, it proves

$$T_{N_0}^{(k)}((1 + \varepsilon)s_*) = 2L_*\varepsilon^{-1/2} + o(\varepsilon^{-1/2}), \qquad L_* \neq 0,$$

and, for every fixed $0 \leq j \leq k$,

$$\sum_{\substack{N > N_0 \\ N \text{ even}}} |T_N^{(j)}((1 + \varepsilon)s_*)| = o(\varepsilon^{-1/2}).$$

The cited LT argument estimates the two local charts and their torus complement, then uses the weighted Stokes current and the Vandermonde and Lie derivative bounds in the particle windows. Its normalized susceptibility $1 - M^2 + 2M^2 \sum_{N \text{ even}, N \geq 2} T_N$ equals (1.16) by (1.12). The HT theorem's first coefficient and differentiated-tail estimates on both boundary halves apply to (1.15). The Gibbs-state and uniform-field identifications on the real intervals come from Section 1.

The geometry on this slice is exact:

$$\mathcal{S}_1 = \{z : |\sinh z| = 1\}, \qquad H_1(y) = \operatorname{arsinh}(\cos y), \quad |y| < \pi/2, \qquad L_1(y) = \operatorname{arsinh}(|\cos y|).$$

The map $z \mapsto \sinh z$ maps the central HT cell biholomorphically onto $|s| < 1$ and maps Ω_L onto $|s| > 1$ as its universal cover, with deck translation by $2\pi\mathrm{i}$. The physical prefactor $\beta = z/(2J_x)$ is defined on the cover. At all nonaxis boundary points the map is locally biholomorphic. The boundary critical points with $\cosh z = 0$ are limits of such points. Dense isotropic obstructions therefore pull back to every point of the two boundaries, including those critical points. The factor $\beta = z/(2J_x)$ is nonzero on the selected boundary family.

Completion of the proof of Theorem 1.1. The unequal-ratio part was proved at the start of this section. At the equal ratio, apply the two normalized isotropic theorems through the exact identities and covering maps above. No limit in λ and no phase duality is used. This proves the theorem for every finite positive ratio on both physical temperature sides. □

Corollary 6.1 (Inverse-temperature barriers)**.** *For each fixed $J_x > 0$, the barriers in the β plane are exactly $\partial\Omega_H/(2J_x)$ and $\partial\Omega_L/(2J_x)$ for their respective physical germs. At $\lambda = 1$ they are the correctly interpreted pullbacks of $|\sinh(2\beta J)| = 1$, with the low-temperature covering specified above.*

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